National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Basal Cell Skin Cancer

Version 2.2026 — March 11, 2026

NCCN.org

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

NCCN Guidelines for Patients® available at www.nccn.org/patients

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

***Jeremy Bordeaux, MD, MPH/Chair ϖ**
Case Comprehensive Cancer Center/
University Hospitals Seidman Cancer Center
and Cleveland Clinic Taussig Cancer Institute

***Paul Nghiem, MD, PhD/Vice Chair ϖ**
Fred Hutchinson Cancer Center

Sumaira Z. Aasi, MD ϖ
Stanford Cancer Institute

Murad Alam, MD, MBA, MSCI ϖ ¶ ζ
Robert H. Lurie Comprehensive Cancer
Center of Northwestern University

Arya Amini, MD §
City of Hope National Medical Center

Diana Bolotin, MD, PhD ϖ
The UChicago Medicine
Comprehensive Cancer Center

Jennifer Chang, MD ф
UC San Diego Moores Cancer Center

Pei-Ling Chen, MD, PhD ≠
Moffitt Cancer Center

Carlo M. Contreras, MD ¶
The Ohio State University Comprehensive
Cancer Center - James Cancer Hospital
and Solove Research Institute

Jessica M. Donigan, MD ϖ ¶
Huntsman Cancer Institute
at the University of Utah

Daniel Eisen, MD ϖ
UC Davis Comprehensive Cancer Center

Jeffrey M. Farma, MD ¶
Fox Chase Cancer Center

Kelly Harms, MD, PhD ϖ
University of Michigan Rogel Cancer Center

Nicole LeBoeuf, MD, MPH ϖ
Dana-Farber/Brigham and Women's Cancer
Center | Mass General Cancer Center

John Nicholas Lukens, MD §
Abramson Cancer Center
at the University of Pennsylvania

Susan Manber ¥
Publicis Health

Lawrence Mark, MD, PhD ϖ
Indiana University Melvin and Bren Simon
Comprehensive Cancer Center

Stacy McMurray, MD ϖ
Vanderbilt-Ingram Cancer Center

Theresa Medina, MD †
University of Colorado Cancer Center

Kishwer S. Nehal, MD ϖ
Memorial Sloan Kettering Cancer Center

Kelly Olino, MD ¶
Yale Cancer Center/Smilow Cancer Hospital

Gyorgy Paragh, MD, PhD, FAAD ϖ
Roswell Park Comprehensive Cancer Center

Soo Park, MD †
UC San Diego Moores Cancer Center

Tejesh Patel, MD ϖ
The University of Tennessee Health Science Center

Jason Rich, MD ζ
Siteman Cancer Center at Barnes-Jewish Hospital
and Washington University School of Medicine

Ashok R. Shaha, MD ¶
Memorial Sloan Kettering Cancer Center

Bhavina Sharma, MD, MPH ‡
Fred & Pamela Buffett Cancer Center

Olayemi Sokumbi, MD ϖ ≠
Mayo Clinic Comprehensive Cancer Center

Divya Srivastava, MD ϖ
UT Southwestern Simmons Comprehensive Cancer Center

Joel Sunshine, MD, PhD ϖ ≠
Johns Hopkins Kimmel Cancer Center

Valencia Thomas, MD ϖ ≠ ¶
The University of Texas MD Anderson Cancer Center

Puja Venkat, MD §
UCLA Jonsson Comprehensive Cancer Center

Melodi Whitley, MD, PhD ϖ
Duke Cancer Institute

Yaohui Gloria Xu, MD, PhD ϖ Ÿ
University of Wisconsin
Carbone Cancer Center

Siegrid Yu, MD ϖ
UCSF Helen Diller Family Comprehensive Cancer Center

***Mehran Yusuf, MD §**
O'Neal Comprehensive Cancer Center at UAB

NCCN
Sara Espinosa, PhD
Beth McCullough, RN, BS

ϖ Dermatology
ф Diagnostic/Interventional radiology
‡ Hematology/Hematology oncology
Þ Internal medicine
† Medical oncology
ζ Otolaryngology
≠ Pathology/Dermatopathology
¥ Patient advocacy
§ Radiotherapy/Radiation oncology
Ÿ Reconstructive surgery
¶ Surgery/Surgical oncology
* Discussion Section Writing Committee

NCCN Guidelines Panel Disclosures

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

NCCN Guidelines Version 2.2026
Basal Cell Skin Cancer

NCCN Basal Cell Skin Cancer Panel Members
Summary of the Guidelines Updates

Clinical Presentation, Workup, Risk Status, and Staging (BCC-1)
Stratification to Determine Treatment Options for Local BCC Based on Risk Factors for Recurrence (BCC-2)
Treatment for Low-Risk BCC (BCC-3)
Treatment for High-Risk BCC (BCC-4)
Treatment for Advanced BCC (BCC-5)
Follow-up and Recurrence (BCC-6)

Principles of Pathology (BCC-A)
Principles of Treatment (BCC-B)
Principles of Radiation Therapy (BCC-C)
Principles of Systemic Therapy (BCC-D)
Principles of Cancer Risk Assessment and Counseling (BCC-E)

Abbreviations (ABBR-1)

Find an NCCN Member Institution: https://www.nccn.org/home/member-institutions.

NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

NCCN Categories of Preference: All recommendations are considered appropriate.

See NCCN Categories of Preference.

The NCCN Guidelines® are a statement of evidence and consensus of the authors regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult the NCCN Guidelines is expected to use independent medical judgment in the context of individual clinical circumstances to determine any patient's care or treatment. The National Comprehensive Cancer Network® (NCCN®) makes no representations or warranties of any kind regarding their content, use or application and disclaims any responsibility for their application or use in any way. The NCCN Guidelines are copyrighted by National Comprehensive Cancer Network®. All rights reserved. The NCCN Guidelines and the illustrations herein may not be reproduced in any form without the express written permission of NCCN. ©2026.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

Updates in Version 2.2026 of the NCCN Guidelines for Basal Cell Skin Caner include:

MS-1
• The discussion section has been updated to reflect changes in the algorithm.

Updates in Version 1.2026 of the NCCN Guidelines for Basal Cell Skin Caner include:

BCC-1
• Preliminary Workup, bullet removed: Shave removal if applicable.
• New footnote c added: Shave removal can be considered for clinically low-risk BCC in low-risk site and in situations where additional treatment would pose morbidity/hardship. (Also page BCC-3A)
BCC-2
• New location of Stratification to Determine Treatment Options for Local BCC Based on Risk Factors for Recurrence.
• Footnote k revised: Having basosquamous, infiltrative, sclerosing/morpheaform, micronodular, and BCC with ~~carcinosarcomatous~~ *sarcomatoid* differentiation features in any portion of the tumor. In some cases, basosquamous tumors may be prognostically similar to squamous cell carcinoma (SCC); clinicopathologic correlation is recommended in these cases to further consider prognostic implication.
BCC-3
• Primary Treatment: fourth bullet revised: ~~Radiation therapy (RT)~~ *Consultation with radiation oncologist* for nonsurgical candidates.
BCC-3A
• New footnote p added: Loranger N, et al. J Am Acad Dermatol 2025;92:610-612.
BCC-4
• Following High-risk BCC, new column added: Multidisciplinary consultation recommended at center with specialized expertise to discuss options.
• Primary Treatment, third bullet revised: *Consultation with radiation oncologi*st ~~For~~ *for* nonsurgical candidates ~~consider multidisciplinary consultation and discussion of definitive RT~~.
• Additional Treatment, top option following Positive margins revised: ~~Multidisciplinary consultation to discuss options:~~ Re-resect, if feasible. . .
BCC-5
• New footnote dd added: In certain circumstances, RT can be combined with Vismodegib. Barker CA, et al. J Clin Oncol 2024;42:2327-2335.
BCC-A
• Principles of Biopsy Reporting, fourth bullet revised: Pathologic report should include histologic subtype and presence and extent of any features that would increase the risk for local recurrence, including invasion of tumor beyond reticular dermis and presence*/caliber* of perineural invasion.
• Principles of Excision Reporting, fourth bullet revised: Minimal reporting elements to be reported for all surgical specimens include histologic subtype of BCC, invasion of tumor beyond deep reticular dermis, presence of perineural invasion (if involving nerve below dermis or if largest nerve involved is ≥0.1 mm in caliber) and ~~angiolymphatic~~ *lymphovascular invasion (LVI)*, and peripheral and deep margin status.
• Footnote a revised: Low-risk histologic subtypes include nodular, superficial, and other non-aggressive growth patterns such as keratotic, infundibulocystic, and fibroepithelioma of Pinkus; high-risk subtypes include basosquamous, infiltrative, sclerosing/morpheaform, micronodular, and BCC with ~~carcinosarcomatous~~ *sarcomatoid* differentiation.
BCC-C
• Significant changes made to Principles of Radiation Therapy section.

Continued

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

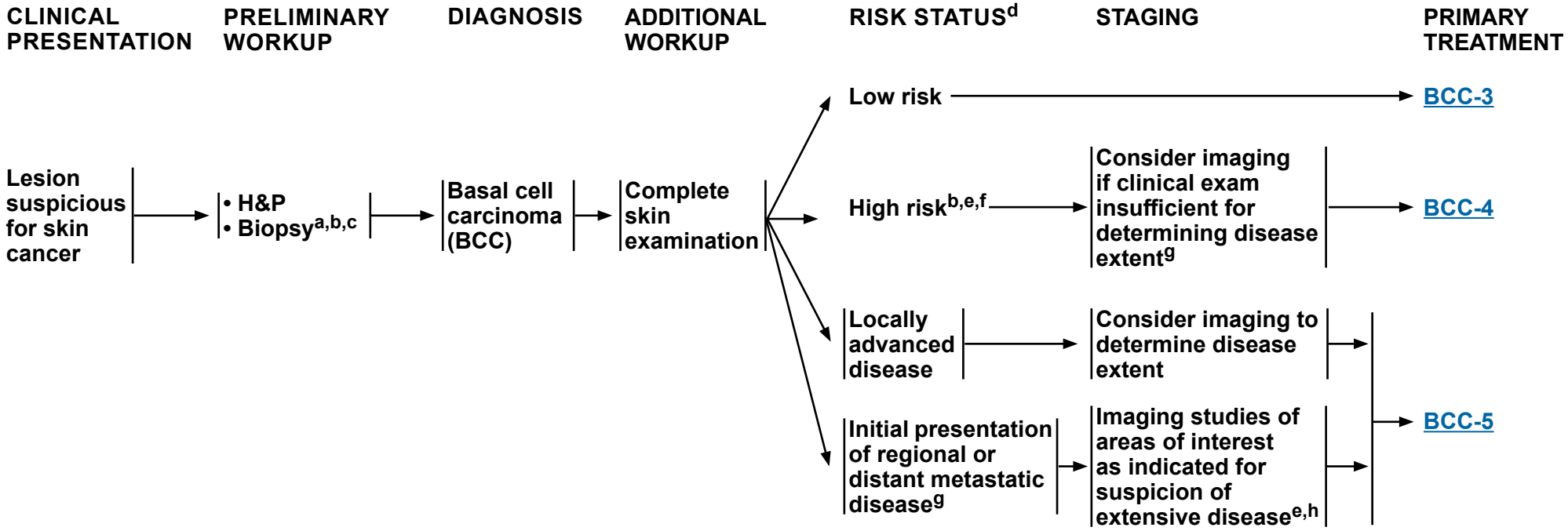
Updates in Version 1.2026 of the NCCN Guidelines for Basal Cell Skin Caner include:

BCC-D 1 of 2
- Metastatic Disease:
  - Preferred, option added: Vismodegib.
  - Other Recommended, option removed: Vismodegib.
- New footnote b added: In certain circumstances, RT can be combined with Vismodegib.

BCC-D 2 of 2
- New reference 7 added: Barker CA, Dufault S, Arron ST, et al. Phase II, single-arm trial of induction and concurrent Vismodegib with curative-intent radiation therapy for locally advanced, unresectable basal cell carcinoma. J Clin Oncol 2024;42:2327-2335.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.



[a] Principles of Pathology (BCC-A).
[b] Risk Factors for Recurrence (BCC-2).
[c] Shave removal can be considered for clinically low-risk BCC in low-risk site and in situations where additional treatment would pose morbidity/hardship.
[d] Morgan FC, et al. J Am Acad Dermatol 2021;85:582-587.
[e] Extensive disease includes deep involvement such as bone, named nerves, and deep soft tissue. If disease of named nerve(s) is suspected, MRI with and without contrast is preferred. If bone disease is suspected, CT with contrast is preferred unless contraindicated.
[f] Any high-risk factor places the patient in the high-risk category.
[g] For rare cases that present with regional or distant metastatic disease at diagnosis, treat per nodal or metastatic pathways on BCC-5.
[h] Imaging modality and targeted area should be at the discretion of the treating team based on the suspected extent of disease (ie, local, regional, metastatic). Histologic confirmation is sufficient to diagnose local recurrence, but MRI with and without contrast can be considered to assess extent of local disease. For nodal or distant metastasis, histologic analysis and/or CT imaging can be used for confirmation and to gauge extent of disease.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

STRATIFICATION TO DETERMINE TREATMENT OPTIONS FOR LOCAL BCC BASED ON RISK FACTORS FOR RECURRENCE[f]

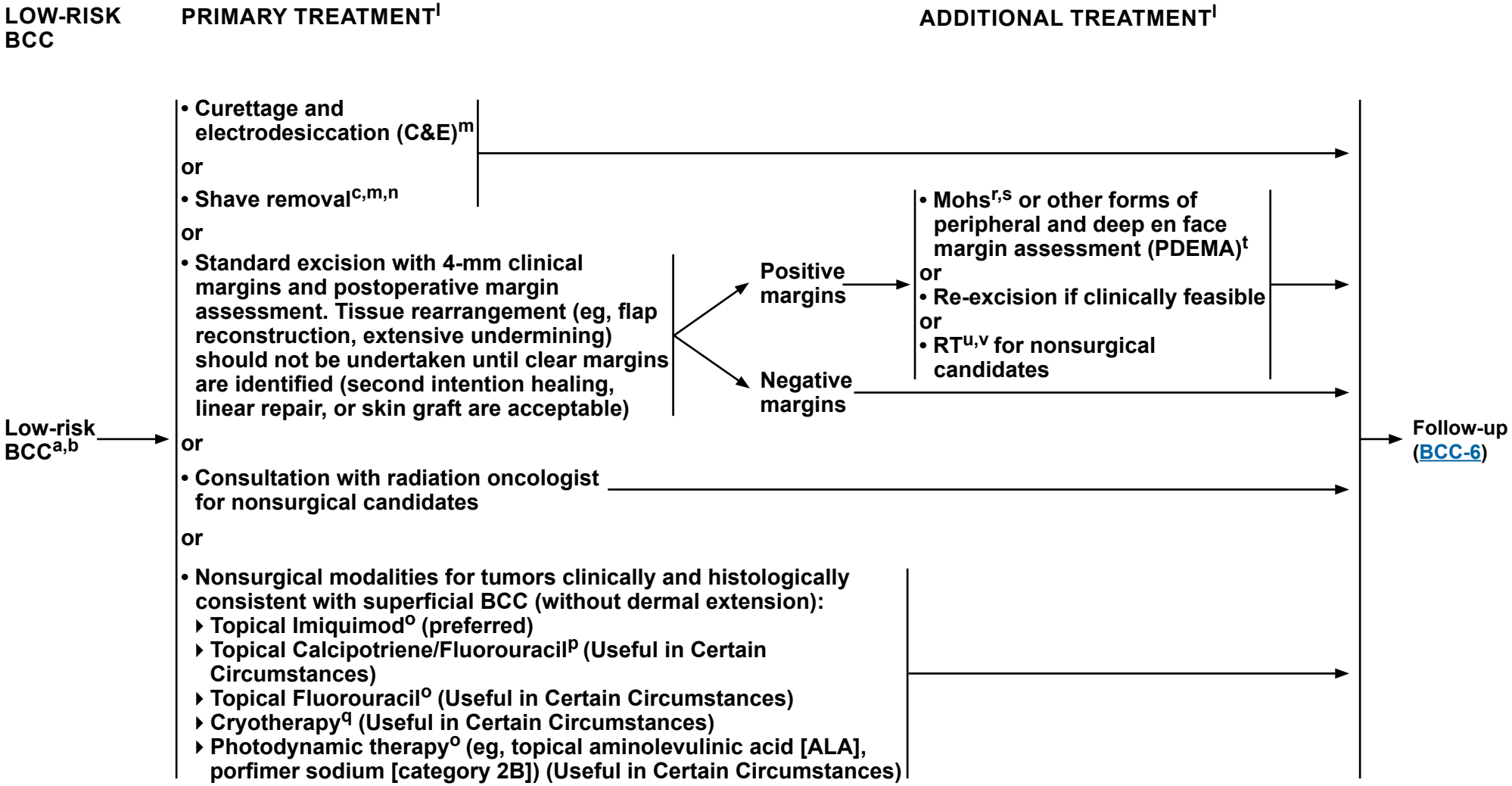
Risk Group	Low Risk	High Risk
Treatment options	BCC-3	BCC-4
H&P		
Location/diameter (cm)	Trunk, extremities <2 cm	Trunk, extremities ≥2 cm
		Head, neck, hands, feet, pretibial, and anogenital area (any size)[j]
Clinical borders	Well-defined	Poorly-defined
Primary vs. recurrent	Primary	Recurrent
Immunosuppression	(-)	(+)
Site of prior RT	(-)	(+)
Pathology (BCC-A)		
Histologic subtype	Nodular, superficial[i]	Aggressive growth pattern[k]
Perineural involvement	(-)	(+)

[f] Any high-risk factor places the patient in the high-risk category.
[i] Low-risk histologic subtypes include nodular, superficial, and other non-aggressive growth patterns such as keratotic, infundibulocystic, and fibroepithelioma of Pinkus.
[j] Narrow excision margins due to anatomic and functional constraints are associated with increased recurrence rates with standard histologic processing. Complete margin assessment with Mohs/PDEMA is recommended. For tumors <6 mm in size, without other high-risk features, other treatment modalities may be considered if at least 4-mm clinically tumor-free margins can be obtained without significant anatomic or functional distortions.
[k] Having basosquamous, infiltrative, sclerosing/morpheaform, micronodular, and BCC with sarcomatoid differentiation features in any portion of the tumor. In some cases, basosquamous tumors may be prognostically similar to squamous cell carcinoma (SCC); clinicopathologic correlation is recommended in these cases to further consider prognostic implication.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.



Footnotes on BCC-3A

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

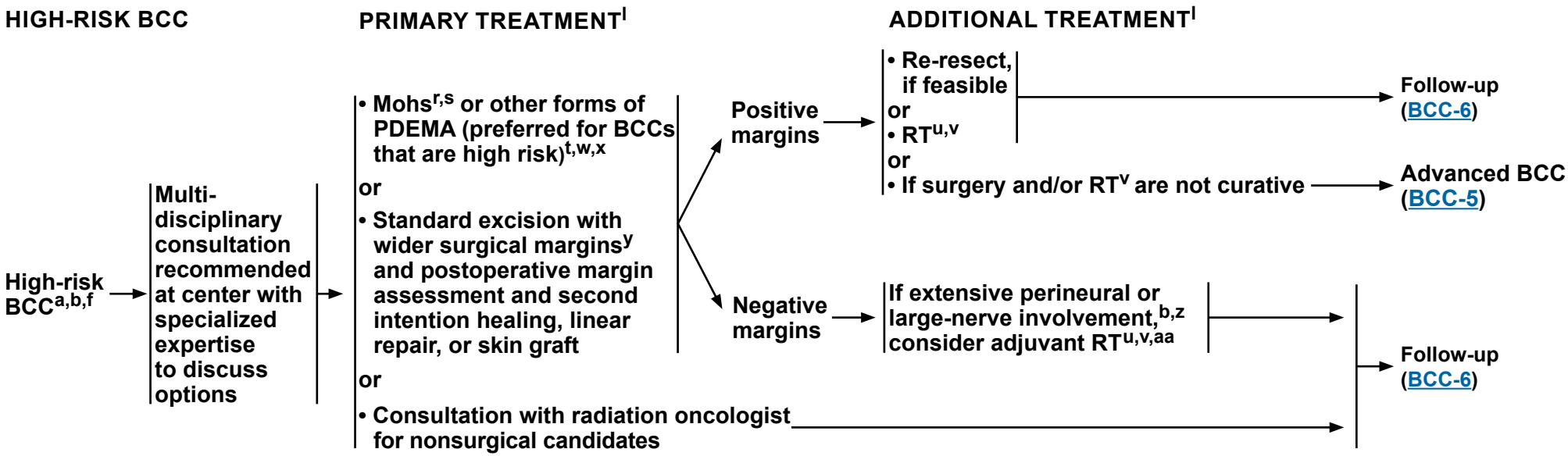
FOOTNOTES

[a] Principles of Pathology (BCC-A).
[b] Risk Factors for Recurrence (BCC-2).
[c] Shave removal can be considered for clinically low-risk BCC in low-risk site and in situations where additional treatment would pose morbidity/hardship.
[l] Principles of Treatment (BCC-B).
[m] If tumor appears to extend beyond the dermis, surgical excision should generally be performed rather than C&E or shave removal.
[n] Shave removal (shaving of epidermal or dermal lesion) is a sharp removal by transverse bowl-shaped slicing to remove epidermal and dermal lesions (without including fat) and does not require suture closure. Emmett AJ, et al. Plast Reconstr Surg 1987;80:47-54; Abramson AK, et al. Dermatol Surg 2013;39:387-392; Wu X, et al. J Am Acad Dermatol 2015;73:791-798; Dando EE, et al. Dermatol Surg 2023;49:130-134.
[o] Cure rates are approximately 10% lower than for surgical treatment modalities. Jansen MHE, et al. J Invest Dermatol 2018;138:527-533; Drew BA, et al. Dermatol Surg 2017;43:1423-1430.
[p] Loranger N, et al. J Am Acad Dermatol 2025;92:610-612.
[q] Current guidelines for low-risk BCC including superficial BCC (sBCC) recommend several treatment options including destructive treatment methods, such as cryosurgery with or without prior curettage or curettage and electrodesiccation. Backman EJ, et al. J Eur Acad Dermatol Venereol 2022;36:1758-1765.
[r] Mohs surgery should be performed by dermatologic surgeons who have specialized training and experience in this procedure.
[s] As per other appropriate use criteria. Task Force/Committee Members; Vidal CI, et al. J Am Acad Dermatol 2019;80:189-207.
[t] PDEMA with permanent section analysis or intraoperative frozen section analysis is an alternative to Mohs. See Principles of PDEMA Technique (SCC-G) within the NCCN Guidelines for Squamous Cell Skin Cancer.
[u] Principles of Radiation Therapy (BCC-C).
[v] Determination of the appropriateness of RT should be performed together with a radiation oncologist.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.



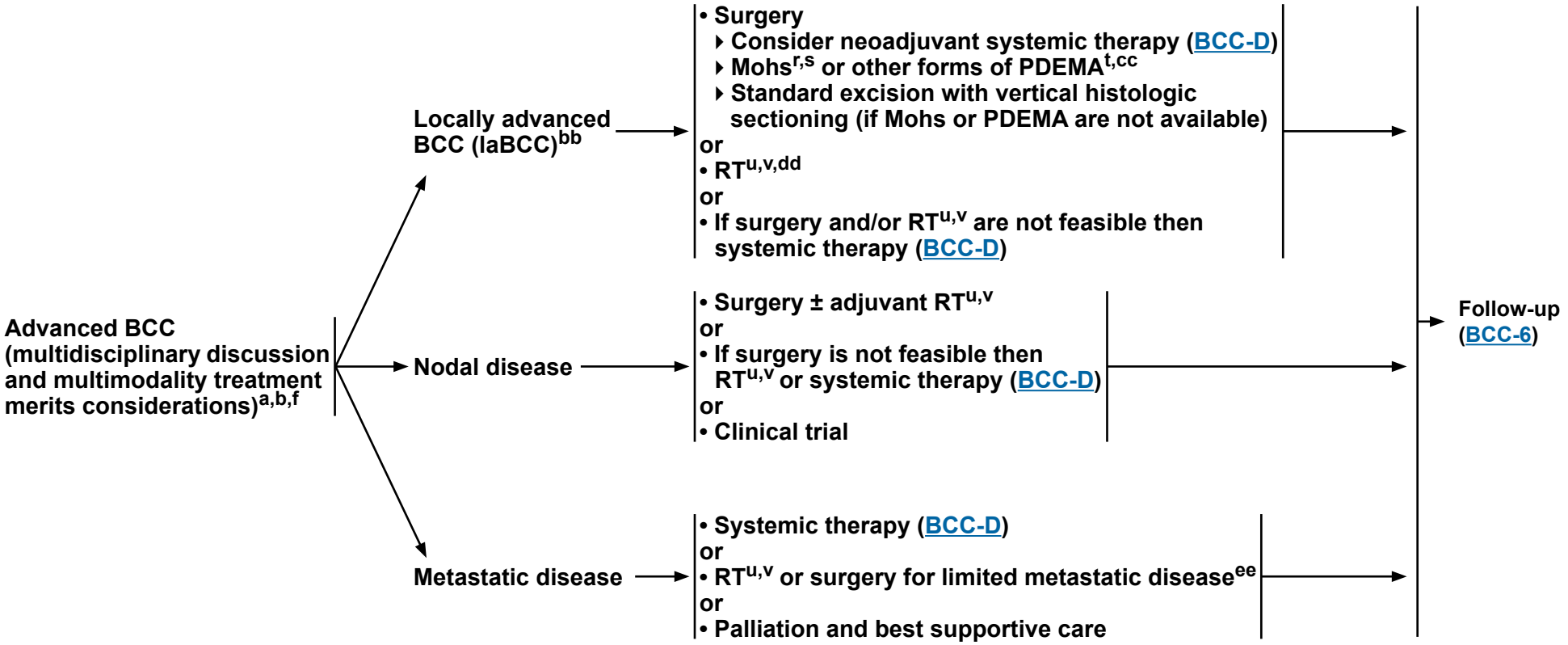
[a] Principles of Pathology (BCC-A).
[b] Risk Factors for Recurrence (BCC-2).
[f] Any high-risk factor places the patient in the high-risk category.
[l] Principles of Treatment (BCC-B).
[r] Mohs surgery should be performed by dermatologic surgeons who have specialized training and experience in this procedure.
[s] As per other appropriate use criteria. Task Force/Committee Members; Vidal CI, et al. J Am Acad Dermatol 2019;80:189-207.
[t] PDEMA with permanent section analysis or intraoperative frozen section analysis is an alternative to Mohs. See Principles of PDEMA Technique (SCC-G) within the NCCN Guidelines for Squamous Cell Skin Cancer.
[u] Principles of Radiation Therapy (BCC-C).
[v] Determination of the appropriateness of RT should be performed together with a radiation oncologist.
[w] For clinically diagnosed non-facial BCCs <6 mm in depth on the head, neck, hands, feet, pretibial, and anogenital area that are clinically confined to the dermis, C&E or shave removal may be considered as an alternative primary treatment option if Mohs, resection with PDEMA, and standard excision are difficult to perform due to patient comorbidities (eg, thrombocytopenia, immunosuppression, bleeding diathesis, multiple primary BCCs). See Risk Factors for Recurrence (BCC-2).
[x] Aggressive histologic subtype is defined as: BCC with squamous differentiation, infiltrative, micronodular, morpheaform, sclerodermiform, or sclerosing. van Loo E, et al. Eur J Cancer 2014;50:3011-3020; Curtis KK, et al. J Natl Compr Canc Netw 2024;22:e247036.
[y] Due to the wide variability of clinical characteristics that may define a high-risk tumor, it is not feasible to recommend a defined margin for standard excision of high-risk BCC. Keen awareness of the subclinical extension of BCC is advised when selecting a treatment modality without complete margin assessment for a high-risk tumor. These margins may need to be modified based on tumor- or patient-specific factors.
[z] If named nerve involvement is suspected, consider MRI with and without contrast of region of interest to evaluate extent and rule out base of skull involvement or intracranial extension in head and neck tumors.
[aa] There are conflicting data about the value of adjuvant RT following margin-negative surgical excision, particularly after Mohs.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

ADVANCED BCC



[a] Principles of Pathology (BCC-A).
[b] Risk Factors for Recurrence (BCC-2).
[f] Any high-risk factor places the patient in the high-risk category.
[l] Principles of Treatment (BCC-B).
[r] Mohs surgery should be performed by dermatologic surgeons who have specialized training and experience in this procedure.
[s] As per other appropriate use criteria. Task Force/Committee Members; Vidal CI, et al. J Am Acad Dermatol 2019;80:189-207.
[t] PDEMA with permanent section analysis or intraoperative frozen section analysis is an alternative to Mohs. See Principles of PDEMA Technique (SCC-G) within the NCCN Guidelines for Squamous Cell Skin Cancer.
[u] Principles of Radiation Therapy (BCC-C).
[v] Determination of the appropriateness of RT should be performed together with a radiation oncologist.
[bb] Primary or recurrent extensive disease where surgery and/or RT may not result in a cure or would possibly produce a significant functional limitation.
[cc] Curtis KK, et al. J Natl Compr Canc Netw 2024;22:e247036.
[dd] In certain circumstances, RT can be combined with Vismodegib. Barker CA, et al. J Clin Oncol 2024;42:2327-2335.
[ee] Under highly selective circumstances, in the context of multidisciplinary consultation, resection of limited metastases can be considered.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

FOLLOW-UP

- H&P
  - Including complete skin exam every 6–12 mo for the first 5 years, and then at least annually for life[ff]
- Consider imaging if clinical exam is insufficient for following the disease[h]
- Patient education:
  - Sun protection
  - Self-examination

RECURRENCE

Local → Follow Primary Treatment pathway for High-risk disease (BCC-4)

Advanced disease:
- Locally advanced
- Nodal metastases
- Distant metastases

→ Follow Primary Treatment pathways for Advanced BCC (BCC-5)

[h] Imaging modality and targeted area should be at the discretion of the treating team based on the suspected extent of disease (ie, local, regional, metastatic). Histologic confirmation is sufficient to diagnose local recurrence, but MRI with and without contrast can be considered to assess extent of local disease. For nodal or distant metastasis, histologic analysis and/or CT imaging can be used for confirmation and to gauge extent of disease.

[ff] Follow-up with a dermatologist is strongly recommended if any of the following criteria are met: past or imminent solid organ, marrow, or hematopoietic cell transplant; one or more cutaneous melanomas in the past 5 years; or four or more non-melanoma skin cancers in the past 5 years.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

PRINCIPLES OF PATHOLOGY

Principles of Biopsy Reporting:

- **The intent of a biopsy is for diagnosis, not to assess the margin status.**
- **Pathologic evaluation of skin biopsies is ideally performed by a dermatologist, pathologist, dermatopathologist, or Mohs surgeon who is experienced in interpreting cutaneous neoplasms.**
- **Clinical information to be submitted on biopsy requisition includes patient age and gender, clinical diameter of lesion, anatomic location, and prior treatment of lesion. Additional helpful features to include are immunosuppression and history of RT.**
- **Pathologic report should include histologic subtype[a] and presence and extent of any features that would increase the risk for local recurrence, including invasion of tumor beyond reticular dermis and presence/caliber of perineural invasion.[1]**

Principles of Excision Reporting:

- **The intent of excision is to clear the tumor and thus margin status needs to be reported.**
- **Saucerization specimens intended for definitive surgical therapy should be labeled as such, as they can be histopathologically difficult to distinguish from shave biopsies but must be evaluated for margin status.**
- **Clinical information to be submitted on excision requisition includes patient age and gender, anatomic location, clinical diameter of lesion, and additional clinical information listed above under Principles of Biopsy Reporting.**
- **Minimal reporting elements to be reported for all surgical specimens include histologic subtype of BCC,[a] invasion of tumor beyond deep reticular dermis, presence of perineural invasion (if involving nerve below dermis or if largest nerve involved is ≥0.1 mm in caliber) and lymphovascular invasion (LVI), and peripheral and deep margin status.**
- **For Mohs excisions, reporting of these elements is also encouraged. Since depth of invasion (in mm) may not be ascertained on tangentially cut Mohs specimens, anatomic level of invasion should be reported. Frozen or permanent section analysis of the clinical tumor specimen may be undertaken if needed for complete reporting of features associated with poor prognosis.[2]**

Footnote

[a] Low-risk histologic subtypes include nodular, superficial, and other non-aggressive growth patterns such as keratotic, infundibulocystic, and fibroepithelioma of Pinkus; high-risk subtypes include basosquamous, infiltrative, sclerosing/morpheaform, micronodular, and BCC with sarcomatoid differentiation.

References

[1] Work Group; Invited Reviewers; Kim JYS, et al. Guidelines of care for the management of basal cell carcinoma. J Am Acad Dermatol 2018;78:540-559.
[2] Morgan FC, Ruiz ES, Karia PS, et al. Brigham and Women's Hospital tumor classification system for basal cell carcinoma identifies patients with risk of metastasis and death. J Am Acad Dermatol 2021;85:582-587.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

PRINCIPLES OF TREATMENT

- **The primary treatment goals of BCC is the complete removal of the tumor and the maximal preservation of function and cosmesis. All treatment decisions should be customized to account for the particular factors present in the individual case and for the patient's preference.**
- **Surgical approaches often offer the most effective and efficient means for accomplishing cure, but considerations of function, cosmesis, and patient preference may lead to choosing RT/topical therapy/systemic therapy as primary treatment in order to achieve optimal overall results.**
- **In certain patients at high risk for multiple primary tumors (eg, basal cell nevus syndrome [Gorlin syndrome], xeroderma pigmentosum, history of RT), increased surveillance and consideration of prophylactic measures may be indicated. Refer patients with suspected basal cell nevus syndrome or xeroderma pigmentosum for genetic evaluation.**
- **In patients with superficial basal cell skin cancer, nonsurgical modalities may be considered. (See BCC-3)**
- **When Mohs[a] with margin assessment is being performed and the preoperative biopsy is considered insufficient for providing all the staging information required to properly treat the tumor, submission of the central specimen for vertical paraffin-embedded permanent sections or documentation of staging parameters in Mohs report is recommended.**
- **Use of nicotinamide may be effective in reducing the development of basal cell skin cancers.[1-4]**

Footnote
[a] Mohs surgery should be performed by dermatologic surgeons who have specialized training and experience in this procedure.

References
[1] Chen AC, Martin AJ, Dalziell RA, et al. A phase II randomized controlled trial of nicotinamide for skin cancer chemoprevention in renal transplant recipients. Br J Dermatol 2016;175:1073-1075.
[2] Chen AC, Martin AJ, Choy B, et al. A phase 3 randomized trial of nicotinamide for skin-cancer chemoprevention. N Engl J Med 2015;373:1618-1626.
[3] Allen NC, Martin AJ, Snaidr VA, et al. Nicotinamide for Skin-Cancer Chemoprevention in Transplant Recipients. N Engl J Med 2023;388:804-812.
[4] Mainville L, Smilga AS, Fortin PR. Effect of Nicotinamide in Skin Cancer and Actinic Keratoses Chemoprophylaxis, and Adverse Effects Related to Nicotinamide: A Systematic Review and Meta-Analysis. J Cutan Med Surg 2022;26:297-308.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

PRINCIPLES OF RADIATION THERAPY

General Principles

- **Refer to the ASTRO Guideline on Definitive and Postoperative Radiation Therapy for Basal and Squamous Cell Cancers of the Skin[1] for general indications and dose recommendations.**
- **Protracted fractionation is associated with improved cosmetic results and should be utilized for poorly vascularized or cartilaginous areas.**
- **RT is contraindicated for genetic conditions predisposing to skin cancer (eg, basal cell nevus syndrome and relatively contraindicated for patients with connective tissue diseases (eg, scleroderma).**
- **Given higher complication rates, reirradiation should not be routinely utilized for recurrent disease within a prior radiation field.**
- **Perineural tumor spread (PNTS), defined as clinically or radiographically apparent macroscopic spread along nerves, is considered higher risk and warrants comprehensive coverage of the involved and potentially interconnected, cranial nerve pathways.[2,3]**
- **Recommendations for external beam RT (EBRT) apply to both photons and electrons.**
  - **Image-guided radiation therapy (IGRT) is considered best practice when treating with intensity-modulated radiation therapy (IMRT), proton beam radiotherapy, or 3D conformal radiation. The use of IGRT for other types of radiotherapy to treat skin cancer is considered unnecessary.[a]**
- **Application of bolus over the primary tumor should be considered to ensure full dose to the surface. Use of surface dosimeters should be considered to verify delivered dose.**
- **Radiation treatments should be given by a practicing radiation oncologist with radiation physics support to meet established quality assurance and dosimetric constraints.**

Primary Tumor and Regional Disease RT Dosing Table on BCC-C (2 of 4)

Brachytherapy RT Dosing Table on BCC-C (3 of 4)

[a] See Discussion.

Note: All recommendations are category 2A unless otherwise indicated.

References on BCC-C (4 of 4)

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

PRINCIPLES OF RADIATION THERAPY
PRIMARY TUMOR AND REGIONAL DISEASE

Primary Tumor	RT Dosing	Common Examples
Adjuvant RT[2]		
• **Conventional fractionation (1.8–2 Gy/fraction)**	BED10 of 60–79 Gy	• 60–66 Gy in 2 Gy/fraction[4] • 60–66 Gy in 1.8 Gy/fraction (reserved for areas with poor wound healing, [eg, lower extremity])
• **Hypofractionation**	BED10 of 48–70 Gy	• 2.5 Gy x 20 fractions[5] • 3 Gy x 15–17 fractions to 45–51 Gy • 6 Gy x 5 fractions (non-consecutive days)
Definitive EBRT		
• **Conventional fractionation (1.8–2 Gy/fraction)**	BED10 of 70–84 Gy	• 60 Gy (for cosmetically sensitive areas) • 66–70 Gy (for locally advanced tumors involving bone or cartilage)
• **Hypofractionation**	BED10 of 48–72 Gy[c]	• Tumors <2 cm: 2.5 Gy x 20 fractions • Tumors ≥2 cm: 2.5–3 Gy x 20–22 fractions • 3 Gy x 17–18 fractions to 51–54 Gy[6] • 4.4 Gy x 10 fractions, 4 fractions per week[7] • 6–7 Gy x 5 fractions, 2 treatments per week[d,7,8]
Regional Disease		
• **Conventional fractionation (1.8–2 Gy/fraction)** ▸ Macroscopic/gross residual lymph nodes ▸ Lymph node + dissected nodal basins ▸ PNTS[b,3]	 66–70 Gy 60 Gy 50–66 Gy	

- **Conventionally fractionated radiotherapy consists of five daily treatments per week.**
- **Hypofractionated radiotherapy consists of fewer treatments with larger fraction size.**

[b] Track involved cranial nerve to base of skull and at-risk nerve interconnections.
[c] Given lack of randomized data, the optimal dose/fractionation is not established, but generally, a goal BED10 of 48–72 Gy is recommended.
[d] Generally reserved for patients who are older or unable to come for protracted course.

References on BCC-C (4 of 4)

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

PRINCIPLES OF RADIATION THERAPY
BRACHYTHERAPY

General Principles – Skin Brachytherapy

- **Isotope-based brachytherapy can be an effective treatment for certain sites of disease, particularly on the head and neck.[a]**
- **There are insufficient long-term efficacy and safety data to support the routine use of electronic surface brachytherapy.[a]**
- **Consider longer courses of RT (ie, 16 fractions) for sensitive areas such as ears, nose, or near eyes.**
- **Consider a higher total dose for patients who are immunocompromised (ie, 3 Gy x 16 fractions).**
- **Consider a lower dose for low-risk, small tumors when using a single channel applicator (ie, 3 Gy x 12 fractions).[9]**
- **Consider BID fractionation if performing interstitial brachytherapy.**
- **Consider shielding normal structures when possible (ie, eye shields, internal nose shields).**
- **Start adjuvant radiation within 4–8 weeks of surgery after incision has healed.**
- **Surface applicators should only be used for superficial tumors (<5 mm depth).**
- **If deeper than 5 mm, consider interstitial brachytherapy or EBRT techniques.**
- **Consider referral to a center with expertise with brachytherapy.**

Primary Tumor	RT Dosing	Common Examples
Definitive or Adjuvant Skin Brachytherapy		
• **Custom surface mold applicators:** Prescription depth ≤5 mm	GOAL BED10 at least 60 Gy	• 3 Gy x 16 fractions (daily) • 5 Gy x 8 fractions (2 times per week [for non-cosmetically sensitive areas])
• **Interstitial brachytherapy:** Prescription depth >5 mm		• 4.4 Gy x 10 fractions (BID [6 hours minimum between fractions])
• **Shielded single channel surface applicators (Leipzig or Valencia):** Prescription depth ≤3 mm		• 3 Gy x 12–16 fractions (daily)

[a] See Discussion.

Note: All recommendations are category 2A unless otherwise indicated.

References on BCC-C (4 of 4)

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

REFERENCES

[1] Likhacheva A, Awan M, Barker CA, et al. Definitive and Postoperative Radiation Therapy for Basal and Squamous Cell Cancers of the Skin: Executive Summary of an American Society for Radiation Oncology Clinical Practice Guideline. Pract Radiat Oncol 2020;10:8-20.
[2] Porceddu SV, Daniels C, Yom SS, et al. Head and Neck Cancer International Group (HNCIG) consensus guidelines for the delivery of postoperative radiation therapy in complex cutaneous Squamous Cell Carcinoma of the Head and Neck (cSCCHN). Int J Radiat Oncol Biol Phys 2020;107:641-651.
[3] Bakst RL, Glastonbury CM, Parvathaneni U, et al. Perineural invasion and perineural tumor spread in head and neck cancer. Int J Radiat Oncol Biol Phys 2019;103:1109-1124.
[4] McDowell L, Yom SS. Locally advanced non-melanomatous skin cancer: Contemporary radiotherapeutic management. Oral Oncol 2019;99:104443.
[5] Hu M, Liu H, Bernard A, et al. Hypofractionated adjuvant radiotherapy in cutaneous squamous cell and basal cell carcinoma of the head and neck: 50(Gy) in 20 study. Oral Oncol Rep 2025;13:100732.
[6] van Hezewijk M, Creutzberg CL, Putter H, et al. Efficacy of a hypofractionated schedule in electron beam radiotherapy for epithelial skin cancer: Analysis of 434 cases. Radiother Oncol 2010;95:245-249.
[7] Veness M. Hypofractionated radiotherapy in older patients with non-melanoma skin cancer: Less is better. Australas J Dermatol 2018;59:124-127.
[8] Dundar Y, Cannon RB, Hunt JP, et al. Radiotherapy regimens in patients with nonmelanoma head and neck skin cancers. Int J Dermatol 2018;57:441-448.
[9] Gauden R, Pracy M, Avery AM, et al. HDR brachytherapy for superficial non-melanoma skin cancers. J Med Imaging Radiat Oncol 2013;57:212-217.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

PRINCIPLES OF SYSTEMIC THERAPY

- **Systemic therapy may be considered for laBCC. Locally advanced disease is defined by those that have primary or recurrent extensive disease where surgery and/or RT may not result in a cure or would possibly produce a significant functional limitation.**
- **Systemic therapy may be considered for cases of nodal or distant metastatic disease, especially if surgery and RT are not feasible.**
- **Multidisciplinary consultation may be required to determine the best treatment approach and deem the tumor not amendable to surgery or RT.**
- **Hedgehog pathway inhibitors (HHIs)**
  - **Due to frequency of intolerable side effects associated with HHIs, drug holidays or other alternatives to daily dosing can be used to reduce side effects to improve adherence to therapy and quality of life.**
  - **HHIs may be considered for diffuse BCC formation (eg, basal cell nevus syndrome or other genetic forms of multiple BCC). HHIs are not FDA approved for basal cell nevus syndrome; however, they may be used off-label and are effective based on a randomized controlled trial.[1]**
- **The role of adjuvant systemic therapy for resected BCC is unclear and thus, adjuvant systemic therapy is best performed in a clinical trial setting.**

Locally Advanced (laBCC), Nodal or Distant Metastatic Basal Cell Carcinoma (mBCC)			
	Preferred	**Other Recommended**	**Useful in Certain Circumstances**
Locally Advanced Disease - Neoadjuvant	• None	• Vismodegib[a,4] (category 2B)	• Cemiplimab-rwlc[c] (category 2B)
Locally Advanced Disease	• None	• Sonidegib[5] • Vismodegib[2,3,b]	• Cemiplimab-rwlc[c,d,6]
Nodal Disease	• None	• Vismodegib • Sonidegib[5] (category 2B)	• Cemiplimab-rwlc[c]
Metastatic Disease	• Vismodegib[2,3]	• None	• Cemiplimab-rwlc[c,6]

[a] In one study of 55 patients with laBCC, neoadjuvant administration of Vismodegib before planned surgery allowed for a smaller surgical procedure in 71% of patients, although it carried a high (36.4%) recurrence risk.[2]
[b] In certain circumstances, RT can be combined with Vismodegib.[7]
[c] Cemiplimab-rwlc is FDA approved for patients with laBCC or mBCC previously treated with an HHI or for whom an HHI is not appropriate.
[d] A multinational single-arm phase 2 trial, consisting of 84 patients with locally advanced BCC (local invasion precluding complete resection or in locations for which surgery may result in severe disfigurement or dysfunction) whose disease had progressed on or was intolerant to prior HHI therapy, was conducted. Thirty-one percent had an objective response, including 6% with a complete response. See Discussion.[6]

References on BCC-D (2 of 2)

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

REFERENCES

[1] Tang JY, Ally MS, Chanana AM, et al. Inhibition of the hedgehog pathway in patients with basal-cell nevus syndrome: final results from the multicentre, randomised, double-blind, placebo-controlled, phase 2 trial. Lancet Oncol 2016;17:1720-1731.
[2] Sekulic A, Migden MR, Oro AE, et al. Efficacy and safety of vismodegib in advanced basal-cell carcinoma. N Engl J Med 2012;366:2171-2179.
[3] Dreno B, Basset-Seguin N, Caro I, et al. Clinical benefit assessment of vismodegib therapy in patients with advanced basal cell carcinoma. Oncologist 2014;19:790-796.
[4] Bertrand N, Guerreschi P, Basset-Seguin N, et al. Vismodegib in neoadjuvant treatment of locally advanced basal cell carcinoma: First results of a multicenter, open-label, phase 2 trial (VISMONEO study): Neoadjuvant Vismodegib in Locally Advanced Basal Cell Carcinoma. eClinicalMedicine 2021;35:100844.
[5] Dummer R, Guminksi A, Gutzmer R, et al. Long-term efficacy and safety of sonidegib in patients with advanced basal cell carcinoma: 42-month analysis of the phase II randomized, double-blind BOLT study. Br J Dermatol 2020;182:1369-1378.
[6] Stratigos AJ, Sekulic A, Peris K, et al. Cemiplimab in locally advanced basal cell carcinoma after hedgehog inhibitor therapy: an open-label, multi-centre, single-arm, phase 2 trial. Lancet Oncol 2021;22:848-857.
[7] Barker CA, Dufault S, Arron ST, et al. Phase II, single-arm trial of induction and concurrent Vismodegib with curative-intent radiation therapy for locally advanced, unresectable basal cell carcinoma. J Clin Oncol 2024;42:2327-2335.

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

PRINCIPLES OF CANCER RISK ASSESSMENT AND COUNSELING

- **The decision to offer genetic testing involves three related stages:**
  1) Pre-test counseling prior to ordering testing;
  2) Consideration of the most appropriate testing strategy; and
  3) Testing result disclosure and post-test counseling.
- **In certain patients at high risk for multiple primary tumors (eg, basal cell nevus syndrome, xeroderma pigmentosum, history of RT), increased surveillance and consideration of prophylactic measures may be indicated. Patients with these conditions should be referred to a cancer center with particular expertise in BCC prevention and prophylaxis.**
- **It is recommended that a genetic counselor, medical geneticist, endocrinologist, oncologist, surgeon, oncology nurse, or other health professional with expertise and experience in cancer genetics be involved at each stage whenever possible. Clinicians without direct referral access to the appropriate expertise should be aware of the telehealth genetic counseling options available. These resources can be found through the National Society of Genetic Counselors (NSGC) "Find a Genetic Counselor" tool (www.nsgc.org).**

See the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate for the following:
- **Principles of Cancer Risk Assessment and Counseling (EVAL-A)**
- **Pedigree: First-, Second-, and Third-Degree Relatives of Proband (EVAL-B)**
- **General Testing Criteria (CRIT-1)**

Note: All recommendations are category 2A unless otherwise indicated.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

ABBREVIATIONS

ALA	aminolevulinic acid
BCC	basal cell carcinoma
BED	biologically effective dose
C&E	curettage and electrodesiccation
EBRT	external beam RT
ENI	elective nodal irradiation
H&P	history and physical
HHI	hedgehog pathway inhibitors
IGRT	image-guided radiation therapy
IMRT	intensity-modulated radiation therapy
laBCC	locally advanced basal cell carcinoma
LVI	lymphovascular invasion
mBCC	metastatic basal cell carcinoma
NSGC	National Society of Genetic Counselors
PDEMA	peripheral and deep en face margin assessment
PNTS	perineural tumor spread
sBCC	superficial basal cell carcinoma
SCC	squamous cell carcinoma

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

Version 2.2026, 03/11/2026 © 2026 National Comprehensive Cancer Network® (NCCN®), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

Discussion

This discussion corresponds to the NCCN Guidelines for Basal Cell Skin Cancer. Last updated: March 11, 2026.

Table of Contents

Overview MS-2
Guidelines Update Methodology MS-2
Literature Search Criteria MS-2
Sensitive/Inclusive Language MS-2
Genetics MS-3
Clinical Presentation and Workup MS-3
Stratification of Local BCC Based on Risk Factors for Recurrence MS-3
- History and Physical Examination MS-4
- Pathology MS-4

Treatment Modalities for BCC MS-5
- Curettage and Electrodesiccation MS-5
- Shave Removal MS-6
- Standard Excision with Postoperative Margin Assessment MS-6
- Mohs Micrographic Surgery and Peripheral and Deep En Face Margin Assessment MS-7
- Radiation Therapy MS-8
- Superficial Therapies MS-9
- Systemic Therapy MS-11

Follow-up MS-13
References MS-16

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

Overview

Basal cell carcinoma (BCC) is the most common cancer in the United States. It is estimated that BCCs occur in 2 million Americans annually, exceeding the incidence of all other cancers combined.[1-3] BCCs are at least two times more common than squamous cell carcinomas (SCCs), the second most common type of skin cancer.[1-6] Furthermore, the incidence of this common malignancy is rising rapidly.[1,3,6,7] Compared with SCC, BCCs are much less likely to metastasize, with a metastatic rate of <0.1%, and thus generally have a good prognosis.[8-10] Although rarely metastatic, BCC can produce substantial local destruction along with disfigurement and may involve extensive areas of soft tissue, cartilage, and bone.

A number of risk factors are associated with the development of BCC. The most recognized environmental carcinogen is sunlight. Evidence reveals that the relationship between sun exposure and BCC is complex, and depends on the timing, pattern, and amount of ultraviolet (UV) radiation.[11-15] Fair skin, red or blond hair, and light eye color are associated with BCC as independent risk factors due to greater susceptibility to UV damage.[13,15-22] BCC risk is increased by both UVA and UVB radiation as well as by ionizing radiation. Radiation therapy (RT) for other conditions, especially at a young age, is also associated with an increased risk for developing BCC.[23-27] Most BCC tumors develop on skin sites exposed to radiation—either from the sun or from therapy.[23-25]

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) are available at www.NCCN.org.

Literature Search Criteria

Prior to the update of this version of the NCCN Guidelines® for Basal Cell Skin Cancer, an electronic search of the PubMed database was performed to obtain key literature published since the previous Guidelines update, using the following search term: basal cell skin carcinoma. The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[28]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase 2; Clinical Trial, Phase 3; Clinical Trial, Phase 4; Guideline; Meta-Analysis; Practice Guideline; Randomized Controlled Trial; Systematic Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[29] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Genetics

Extensive research has led to advances in the understanding of the genetics of BCC. The sonic hedgehog signaling pathway has emerged as playing a pivotal role in the pathogenesis of BCC, and mutations in a number of molecules in this pathway have been implicated in the development of the disease.[30-32] Mutations in the *PTCH1* (patched 1) gene on chromosome 9q, which codes for the sonic hedgehog receptor, are the underlying cause of nevoid BCC syndrome, and are present in approximately 30% to 90% of sporadic BCCs.[33-41] Specific UV-induced mutations in the tumor suppressor gene *p53* appear to be a common event in BCC development.[35,38,41,42] Certain genetic syndromes greatly predispose affected individuals to skin cancer formation, including BCC, such as albinism[43,44] and xeroderma pigmentosum (in which defects exist in UV light-induced unscheduled DNA repair).[45-51]

Clinical Presentation and Workup

On clinical presentation of the patient with lesion suspicious of skin cancer, workup for BCC begins with a history and physical examination. Workup also includes skin biopsy on any suspicious lesion. The biopsy should include deep reticular dermis. This procedure is preferred because an infiltrative histology may sometimes be present only at the deeper, advancing margins of a tumor, and superficial biopsies will frequently miss this component.[52,53] Shave removal can be considered for clinically low-risk BCC in a low-risk site and in situations where additional treatment would detrimentally affect morbidity or cause hardship. After BCC diagnosis, a full skin examination is recommended, because individuals with skin cancer often have additional, concurrent precancers or cancers located at other, usually sun-exposed skin sites. These individuals are also at increased risk of developing cutaneous melanoma.[54]

Stratification of Local BCC Based on Risk Factors for Recurrence

After the complete skin examination, a risk assessment should be performed to determine the treatment plan.[55] The NCCN Panel examined risk factors for BCC associated with recurrence (see *Stratification to Determine Treatment Options for Local BCC Based on Risk Factors for Recurrence* in the algorithm). Any high-risk factor places the skin lesion in the high-risk category and imaging should be considered if a clinical exam is insufficient to determine disease extent. Skin lesions in populations placed at increased risk may be difficult to assess clinically; therefore, a low threshold for performing skin biopsies in these patients is necessary. Patients with locally advanced disease, which is defined as primary or recurrent extensive disease where surgery and/or RT may not result in a cure or would potentially yield a significant functional limitation, should consider imaging to determine disease extent. For rare cases when patients present with regional or distant metastatic disease at diagnosis, imaging of areas of interest can be performed when there is suspicion of extensive disease prior to treatment as nodal or distant metastases. Imaging studies may be clinically evident when extensive disease, such as bone involvement, perineural invasion (PNI), or deep soft tissue involvement, is suspected. If perineural disease is suspected, MRI with or without contrast is preferred.[56,57] If bone disease is suspected, CT with contrast is preferred unless contraindicated. Imaging modality and targeted area should be at the discretion of the treating team based on the suspected extent of disease (ie, local, regional, metastatic). Histologic confirmation is sufficient to diagnose local recurrence, but MRI can be considered to assess extent of local disease. For nodal or distant

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

metastases, histologic analysis and/or CT imaging can be used for confirmation and to gauge the extent of disease.

History and Physical Examination

Location and Diameter

Anatomic location[58-64] and diameter[60-66] have been known to be a risk factor for BCC recurrence and metastasis for many years. Historically, anatomical areas have been filtered based on the risk of recurrence for high (H-zone), medium (M-zone), and low risk (L-zone).[67-70] Based on a 27-year retrospective review of 5755 BCCs, recurrences were significantly more common when tumors in high-risk locations (H-zone: central face, eyebrows, nose, lips, chin, ear, temple, genitalia, nipples/areola, hands, feet, ankles, and nail units) were ≥6 mm in diameter and when tumors in moderate-risk locations (M-zone: cheeks, forehead, scalp, neck, jawline, and pretibial surface) were ≥10 mm in diameter.[71] In general, BCCs that develop in the head and neck area, which includes the H-zone of the face, are more likely to recur than those that develop on the trunk and extremities (L-zone). The American Academy of Dermatology in collaboration with American College of Mohs Surgery, American Society for Dermatologic Surgery, and American Society for Mohs Surgery developed an appropriate use criteria document in the treatment of cutaneous neoplasms based on 270 clinical scenarios including 69 BCCs,[72] which has been incorporated into *Stratification to Determine Treatment Options for Local BCC Based on Risk Factors for Recurrence* within the algorithm.

Clinical Borders and Primary Versus Recurrent Disease

The low- and high-risk factors of well-defined versus ill-defined clinical tumor borders[73-75] and primary versus recurrent disease,[62,74,76] respectively, have been extensively documented in the literature.

Immunosuppression

Settings of immunosuppression, such as organ transplantation,[77-82] and long-term use of psoralen and UVA (PUVA) light,[83,84] increase the incidence of BCC. In particular, among patients who have had organ transplants, BCC incidence is approximately 5- to 10-fold higher than in the general population,[85-87] occurring in up to half of patients during the 10 years following transplant.[88-91] Several large retrospective studies found that BCCs in patients who had received organ transplants were more likely to have the superficial histologic subtype and to occur in extracephalic locations and in younger patients (mean age of onset 15 years lower).[92-94] Two of these studies showed similar low recurrence rates for transplant recipients and controls.[93,94] Nevertheless, because of NCCN Guidelines Panel Members' own anecdotal experiences, the Panel decided to classify BCCs developing in settings of immunosuppression as potentially high-risk tumors.

Site of Prior Radiotherapy

Tumors developing in sites of prior RT refer to primary BCCs arising in areas previously irradiated for unrelated conditions. All recurrent tumors, irrespective of prior therapy, are defined as high risk. Data from a number of studies with large sample sizes support that prior RT for unrelated, frequently benign conditions is a risk factor for BCC development.[23-27,95,96]

Pathology

Pathologic Subtypes

Histologic subtyping of BCC as a predictor of risk of recurrence is a well-established concept.[97,98] The subtypes encompassed by the term "aggressive growth pattern," including basosquamous, micronodular, infiltrative, sclerosing, and morpheaform patterns or BCC with sarcomatoid differentiation features, are more likely to recur than the nodular and superficial BCC (sBCC).[65,73,74,76,99-103] Non-aggressive subtypes include the keratotic variant, infundibulocystic variant, and fibroepithelioma of Pinkus.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

Basosquamous carcinomas are tumors that have the histologic appearance of both a BCC and an SCC. Some basosquamous tumors are the result of a BCC colliding with an adjacent SCC. Others represent truly biphenotypic tumors, many of which may have started as BCC, but have subsequently undergone prominent partial squamous metaplasia.[104] Data suggest that basosquamous carcinomas have a metastatic capacity that is more similar to that of SCC than BCC.[105-107]

Perineural Involvement

PNI is uncommon in any nonmelanoma skin cancer (NMSC) (2%–6%), and develops less frequently and is less aggressive in BCC versus SCC.[108-113] BCC with PNI poses a greatly increased risk of recurrence, and is associated with other risk factors including previous recurrent tumors, high grade, larger lesion size, and certain subtypes (infiltrating, morpheaform, and basosquamous).[112,114,115] If large nerve involvement is suspected, MRI should be considered to evaluate extent and/or rule out skull involvement in those with head and neck tumors.[57,116-118] Additionally, in the presence of large caliber or extensive small caliber PNI, a thorough cranial nerve exam is indicated.

Age and Its Effect on BCC Behavior

Whether young age (typically ≤40 years) is an independent risk factor for aggressive BCC behavior is debatable. An analysis of a large database of patients with BCC (N = 3381) documented an increased percentage of BCC with aggressive histologic growth patterns in young persons.[119] In contrast, results from other analyses of large databases (N = 1000 to >10,000) indicate that patients presenting with BCC at a young age are more likely to have the superficial subtype.[120-123] Other analyses report no significant differences in BCC histologic subtype between younger versus older patients.[124-126] The relationship between tumor location and patient age is also unclear, as several studies showed that younger patients were more likely to present with BCCs on the trunk or extremities,[120,125,127,128] while another found no significant association.[124]

Most large studies (N = 50–2000) have shown no significant association between age and recurrence rate.[62,74,124,126] One multivariate analysis, however, showed a positive relationship between increasing age and likelihood of recurrence.[129] Age has also been evaluated as a risk factor for developing a second or multiple BCCs, and many of these studies using fairly large databases (N = 200–2500) found that the risk of developing more than one BCC is associated with increased age.[65,126,128-134] On the contrary, an analysis of a very large database (N = 71,924) found a significantly higher risk of subsequent NMSC in patients <40 years of age at the time of their first BCC diagnosis.[135] In addition, an analysis of 100 metastatic BCCs (mBCCs) found that patients with distant metastases tended to be younger than those with only regional metastases.[136] Consistent with this idea, the Rotterdam Study showed that while the risk of developing a second BCC increased with age,[134] the risk of developing multiple BCC lesions was highest in patients who were <65 years of age at the time of their first BCC diagnosis.[137] Taken together, these studies suggest that young age, in and of itself, is not considered a risk factor for aggressive BCC. Nevertheless, there is a small subset of patients who develop BCC at a young age and may have particularly aggressive disease. These patients may benefit from regular follow-up.

Treatment Modalities for BCC

Curettage and Electrodesiccation

Although a fast and cost-effective technique for superficial lesions, curettage and electrodesiccation (C&E) does not allow histologic margin assessment. Studies have reported overall 5-year recurrence rates ranging from 1.2% to 40% in patients with BCC selected for C&E, with high-risk locations and histologically aggressive subtypes reporting higher recurrence rates.[60,69,138-146]

This technique is deemed effective for properly selected, low-risk BCC with three caveats.[60,144] First, C&E should not be used to treat areas with

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

terminal hair growth such as the scalp, pubic and axillary regions, or beard area due to the risk that a tumor extending down follicular structures might not be adequately removed. Second, if the subcutaneous layer is reached during the course of C&E, then surgical removal should generally be performed instead. This change in therapy is necessary as the effectiveness of the C&E technique rests on the ability of the clinician to distinguish between firm, normal dermis, and soft tumor tissue when using a sharp curette. Since subcutaneous adipose is even softer than tumor tissue, the ability of the curette to distinguish and, therefore, selectively and completely remove tumor cells diminishes. Third, if C&E has been performed based only on the appearance of a low-risk tumor, biopsy results of the tissue taken at the time of C&E should be reviewed to make sure that there are no high-risk pathologic features that would require additional therapy. For tumors on the cheeks, forehead, scalp, neck, and pretibial that are <6 mm in depth and confined to the dermis, C&E may be considered as an alternative primary treatment option if Mohs micrographic surgery or resection with peripheral and deep en face margin assessment (PDEMA) and standard excision are not feasible due to patient comorbidities.

Shave Removal

Shave removal, the shaving of epidermal or dermal lesions, is a sharp removal by bowl-shaped slicing of the epidermal and dermal lesions, without including fat, and does not require suture closure.[147] Like C&E, there is concern for inaccurate margin status assessment with shave removal.[148] However, it can be considered for clinically low-risk BCC in a low-risk site and in situations where additional treatment would detrimentally affect morbidity or cause hardship. Shave removal studies have reported a 0.5% to 30% rate of recurrence over a 3- to 5-year follow-up, multiple tumors treated in single visits, and a risk for misdiagnosis of only 1%.[147-150]

Standard Excision with Postoperative Margin Assessment

Another therapeutic option for BCC is standard surgical excision followed by postoperative pathologic evaluation of margins. This technique has been reported to achieve 5-year recurrence rates of 0.8% to 17.4% for BCC, with lower recurrence rates associated with low-risk tumors and higher recurrence rates associated with high-risk tumors.[138,140,146,151-153] Studies have reported variable margins required to completely excise 95% of all tumor.[154-159] These margins have been suggested to be 2 to 4 mm for low-risk, well-demarcated tumors <2 cm,[154-158] whereas margins of 4 to 6 mm,[155] and in one study, 8 mm[154] were suggested for high-risk BCC. Given this wide variability, studies have reported incomplete excision rates after standard excision ranging from 3.2% to 61.5% depending on tumor location, histologic subtype, and medical provider's specialty.[160-169] Therefore, postoperative margin assessment and identification of clear margins are critical to ensure favorable outcomes with standard excision.

The clinical margins chosen by the Panel for the primary treatment of low-risk BCC are based on the work of Zitelli et al.[170] Their analysis indicated that for well-circumscribed BCC lesions <2 cm in diameter, excision with 4-mm clinical margins should result in complete removal in >95% of cases. The indications for this approach were also expanded to include re-excision of low-risk primary BCC if positive margins are obtained after an initial excision with postoperative margin assessment. For high-risk BCC, standard excision with wider surgical margins is recommended as the primary treatment. Due to the wide variability of clinical characteristics that may define a high-risk tumor, it is not feasible to recommend a defined margin for standard excision of high-risk BCC. Kean awareness of the subclinical extension of BCC is advised when selecting a treatment modality without complete margin assessment for a high-risk tumor. These margins may need to be modified based on tumor- or patient-specific factors. When standard excision with wider surgical margins yields positive

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

margins, Mohs or other forms of PDEMA or standard re-excision are recommended (if PDEMA is not feasible).

For either low-risk or high-risk BCC, when standard excision is used, tissue rearrangement (eg, flap reconstruction, extensive undermining) should not be undertaken until clear margins are identified. Second, intention healing, linear repair, or skin graft are acceptable options.

Mohs Micrographic Surgery and Peripheral and Deep En Face Margin Assessment

PDEMA, or complete margin assessment, is a term used for a subset of surgical techniques for high-quality histologic visualization and interpretation of the margin surface or surgically excised tissue. Mohs micrographic surgery is the most common utilized PDEMA technique. Mohs procedures are particularly successful in non-metastatic basal and squamous cell skin cancers where tissue-sparing and precision microscopic control of margins is a priority and has been associated with lower recurrence rates.[171,172] While PDEMA is a team procedure that requires the participation of physicians from multiple disciplines, Mohs physicians serve as both the surgeon and pathologist requiring highly specialized training. Efforts have been extended to generate consensus recommendations to offer Mohs surgeons guidance and promote standardization to make data aggregate from multicenter clinical trials possible.[173]

Mohs is the preferred surgical technique over standard excision for re-excision of low-risk BCC after positive margins with standard excision, as well as the primary surgical technique of choice for high-risk BCC because it allows intraoperative analysis of 100% of the excision margin. Mohs is also recommended when standard excision with wider surgical margins is unable to achieve negative margins in high-risk BCC. Two meta-analyses published in 1989 associated Mohs with 5-year recurrence rates of 1.0% for primary BCC, and 5.6% for recurrent BCC.[138,146] In these studies, the recurrence rates for Mohs were lower than those for standard excision (10.1% and 17.4% for primary and recurrent BCC, respectively), and lower than those for any other treatment modality included in the analysis (C&E, cryotherapy, and RT).[138,146] Studies on the long-term outcomes (~4 years) of Mohs have reported overall recurrence rates of 2.9% to 3.8%,[174,175] specifically 0% to 6.5% for primary and 4% to 20% for recurrent BCCs.[98,176-181]

A prospective randomized trial comparing Mohs to standard excision reported fewer 10-year recurrences with Mohs for both primary (2.5% vs. 4.1%; $P = .397$) and recurrent BCC (2.4% vs. 12.1%; $P = .015$), although the difference was only statistically significant for recurrent tumors. Importantly, a large proportion of recurrences occurred >5 years after treatment.[69,182,183] It has been demonstrated that H-zone location, recurrent tumor, aggressive subtype, PNI, and tumor size ≥11 mm are significantly associated with two or more Mohs stages.[114,184] However, sBCC, despite being generally considered less aggressive, was shown in a Brazilian study to be 9.03 times more likely to require more than one Mohs stage, likely due to “skip areas” and clinically indistinct borders.[185]

Excision with PDEMA with permanent section analysis or intraoperative frozen section analysis is an acceptable alternative to Mohs provided it includes a complete assessment of all deep and peripheral margins. A 5-year recurrence rate of 0.58% has been reported with slow Mohs using formalin-fixed paraffin-embedded sections and delayed closure in a UK-based prospective study.[186] The descriptive term PDEMA underscores the Panel’s belief that complete histologic assessment of the entire marginal surface is the key to optimal tumor removal. For more information, refer to *Principles of PDEMA Technique* in the NCCN Guidelines for Squamous Cell Skin Cancer, available at www.NCCN.org.

Version 2.2026 © 2026 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

Radiation Therapy

Although surgery is the mainstay of local treatment for BCC, consideration of function, patient preference, and other factors may lead to the choice of RT as primary therapy for non-surgical candidates for both low-risk and high-risk as well as patients with advanced BCC (locally advanced, nodal, and mBCC).[187]

If resection is no longer feasible to obtain negative margins, RT can be considered as an additional treatment option after multidisciplinary consultation for low-risk and high-risk BCC with positive margins. For patients with high-risk BCC and negative margins after resection, adjuvant RT is also a recommended treatment option if extensive perineural or large-nerve involvement is observed.[188] In these patients, local control has been reported to be 50% to 90% with postoperative RT.[187,189] There are conflicting data about the value of adjuvant RT following margin-negative surgical excision, particularly after Mohs. For patients with locally advanced BCC (laBCC) or metastatic disease, RT is recommended as an option for primary treatment. For patients with unresectable laBCC and eligible for RT a combination of vismodegib with RT may be a reasonable approach. RT is also a primary treatment option in instances where surgery is not feasible or as an adjuvant RT combination with surgery.

Two meta-analyses reported 5-year recurrence rates of 8.7% and 9.8% after RT on primary and recurrent BCC, respectively.[138,146] Retrospective analyses of BCC treated with RT have reported 5-year local control, cure, or complete response rates ranging from 93% to 96%,[190-193] and 5-year recurrence rates from 4% to 16%.[194-196] Efficacy of RT was better for BCCs that were less advanced, primary (vs. recurrent), or had smaller diameter or nodular histologic subtype.[190,191,193-195] A prospective study randomizing 347 patients to receive either surgery (standard excision with free margins ≥2 mm from visible borders) or RT as primary treatment of BCC reported higher recurrence rates with RT than surgery (7.5% vs. 0.7%; $P = .003$),[197] poorer cosmetic outcomes, and more postoperative complications.[198]

A small number of prospective studies have reported high rates of tumor control with specific radiation dose fractionation regimens for small BCCs.[197,199,200] A systematic review and meta-analysis also reported hypofractionated RT regimens associated with positive cosmetic outcomes.[201] The NCCN Panel's recommendations for RT dosing are based on clinical practice at their institutions and clinical evidence from studies of various skin cancers. These recommendations are separated into primary tumor (adjuvant RT and definitive EBRT), regional disease, and regarding brachytherapy. In some instances, dose and fractionation schema for both conventional fractionation and hypofractionation is provided.[202,203] Hypofractionation consists of a higher dose of radiation per fraction leading to fewer fractions overall and often a shorter time frame for treatment. For general recommendations regarding radiation doses, refer to the ASTRO Guideline on Definitive and Postoperative Radiation Therapy for Basal and Squamous Cell Cancers of the Skin.[204]

When treating BCC with intensity-modulated RT (IMRT), proton beam RT, or 3D conformal RT (3D-CRT), it is considered best practice to use image-guided RT (IGRT). IGRT for other types of radiation techniques used to treat BCC is considered unnecessary. Tissue equivalent bolus should be considered for photon/electron treatments when therapeutic targets include the skin surface with use of surface dosimeters to confirm delivered dose at skin surface. Proton radiation may be able to treat superficial targets at therapeutic dose without addition of bolus, but discussion with physics/dosimetry is recommended for individual cases. The appropriateness of RT should be determined by a radiation oncologist with radiation physics support to meet established quality assurance and dosimetric constraints. Isotope-based brachytherapy can be an effective treatment for certain sites of disease, particularly on the head and neck.[205-208] Considerations for brachytherapy include sensitive treatment areas (eg,

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

ear, nose, near eyes), if immunocompromised, tumor size,[209] timing of surgical procedures, and optimal brachytherapy technique based on individual patient factors. The NCCN Panel recommendations based on their clinical expertise and available data include definitive or adjuvant skin brachytherapy dosing examples for custom surface mold applicators, interstitial brachytherapy, and shielded single-channel surface applications.

However, there are insufficient long-term efficacy and safety data to support the routine use of electronic surface brachytherapy.[210,211] Electronic brachytherapy (EB) uses electrically generated x-rays, which are not regulated by the Nuclear Regulatory Commission. One of the apparent advantages of EB devices is the decreased shielding requirements and the portability of the units; however, this may lead to EB use in settings unfamiliar with the hazards of therapeutic radiation delivery.[212] The American Association of Physicists in Medicine (AAPM) task group 152 has provided guidelines for proper and safe EB use including having an authorized medical physicist (AMP) physically present from the initiation through the duration of all treatments involving an EB unit. The AMP is responsible for output calibration, quality assurance, training, and treatment planning.[212] Furthermore, training and educational requirements for those administering EB vary considerably from state to state, and users should consult their local radiation safety committee to confirm appropriate adherence. The American Brachytherapy Society (ABS) has published a consensus statement for EB, noting a paucity of long-term clinical outcome data and lack of comparison to surgery or standard radiotherapy techniques.[213] Particularly concerning is the absence of standardized dosimetry (in comparison to high-dose-rate [HDR] brachytherapy and external beam RT [EBRT]) to ensure adequate target coverage, skin surface dose, and plan quality assurance. The ABS, therefore, recommends that “EB treatment should be performed on clinical registry or trial at this time.” The American Academy of Dermatology (AAD) has published a Position Statement: “The AAD believes additional research is needed on electronic surface brachytherapy particularly on long-term outcomes.” The AAD acknowledges concerns regarding aggressive marketing to dermatologists with a focus on revenue streams, potentially undermining quality of care and patient safety. The AAD also cautions dermatologists to be mindful of the Stark physician self-referral law. Recently, image guidance Current Procedural Terminology (CPT) codes have been used for the delivery of EB, despite a lack of evidence demonstrating a benefit in clinical outcomes with image guidance. We do not support the use of image guidance with EB currently. In conclusion, EB should only be performed by radiation oncologists with medical physics support, on registry or clinical trial, with appropriate safeguards, and in adherence with all state and federal regulations. Further research is needed regarding dose deposition and long-term clinical outcomes.

Superficial Therapies

In patients with sBCC, therapies such as topical imiquimod, topical fluorouracil, or photodynamic therapy (PDT) may be considered, although cure rates are approximately 10% lower than for surgical treatment modalities.[214-216] Additional options for patients with sBCC include cryotherapy and topical calcipotriene/fluorouracil.[217,218] These options are also recommended for patients where surgery or RT is contraindicated or impractical and are considered as useful in certain circumstances.

Topical Therapies

Imiquimod was found to be effective for treating nodular and sBCC in randomized studies.[219-224] Two 5-year follow-up studies reported overall treatment success rates of 80.4% and 77.9%, respectively, in patients with sBCC treated with imiquimod.[223,225] Recurrence seems to be associated with tumor thickness.[226] A phase III randomized trial in patients with superficial or nodular BCC showed that imiquimod provided an 82.5%

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

clinical success rate.[227,228] For all of these studies, tumors in the H-zone were excluded. Although the clinical success rate was significantly higher with surgical excision using a 4-mm margin (97.7%; $P < .001$), cosmetic outcomes by dermatologic assessment were significantly better with imiquimod (excellent/good at 3-year follow-up: 61% vs. 36%; $P < .001$).

Another topical cream with efficacy against BCC is fluorouracil,[229,230] which has been shown in a large randomized trial to have a 5-year tumor-free survival probability of 70%.[215,231,232] Other studies have reported cure rates of up to 90% with this treatment.[233-235] A retrospective assessment of response and recurrence rates by Loranger et al reported that the combined topical calcipotriene and fluorouracil effectively treated sBCCs. Overall, 209 tumors were included from the 163 patients in the report where 30 tumors were sBCC and 179 were squamous cell carcinomas in situ (SCCIS). A complete response was achieved in 83% of sBCC tumors (25/30) versus 92% in SCCIS (165/179). The two-year recurrence free survival for sBCCs was 84%. Additionally, it was noted that local recurrence after complete response was more likely to occur in sBCCs (12%) compared to SCCIS (4%). While these are promising results for the treatment of sBCC tumors, further studies are needed to optimally determine specific treatment guidelines with this combination topical regimen.[218]

Photodynamic Therapy

PDT with photosensitizing agents including 5-aminolevulinic acid (ALA) and porfimer sodium is another option for sBCC.[236-238] Multiple randomized trials and a meta-analysis have shown that rates of excellent or good cosmetic outcomes were higher with PDT versus surgery, although surgery was superior to PDT in terms of disease control.[152,239-246] Data from clinical trials reported cure rates from 60% to 100% by PDT for patients with BCC.[242,247-252] Most of these studies have focused on the superficial and nodular histologic subtypes, and several have found higher cure rates for superficial versus nodular subtypes in both low- and high-risk locations.[242,247,252] Ulceration and thickness are associated with lower response to therapy,[252] and within the nodular subtype, cure rates are better with thinner lesions.[241] Clinical studies have demonstrated PDT activity against “difficult-to-treat” lesions, with a 24-month complete response rate of 78%.[247,253] Currently, PDT is being used at some NCCN Member Institutions for premalignant or superficial low-risk lesions on any location on the body, although response rates may be higher on the face and scalp.[254,255]

Cryotherapy

Cryotherapy has been used for many years as a fast and cost-effective means for removal of BCCs.[217] Systematic reviews of historical data in primary BCCs have reported recurrence rates for cryotherapy ranging from 0% to 13%, and mean recurrence rates from pooled analyses ranging between 3% and 4%.[138,140,256,257] In prospective trials, cryotherapy has been shown to result in recurrence rates ranging from 5% to 39%.[199,258-260] Current guidelines for low-risk BCC, which include sBCC, recommend several treatment options such as cryosurgery with or without prior curettage or C&E.[217] A key limitation of cryotherapy is poorer cosmetic outcomes compared with other treatment options, as demonstrated by prospective randomized trials.[259-261]

Comparisons of Superficial Therapies

Several randomized studies and meta-analyses have compared superficial therapies for BCC (Table 1). In summary, these studies indicate that in patients with sBCC, PDT has similar efficacy as cryotherapy but much better cosmetic outcomes. Whereas a meta-analysis of 23 randomized and non-randomized trials found no significant difference in efficacy for PDT versus imiquimod,[262] a randomized trial showed that treatment success was more likely with imiquimod.[215,232] This study also demonstrates superior imiquimod outcomes compared to fluorouracil cream. Exploratory subanalyses found that treatment success rates were significantly higher with imiquimod for tumors that are large or truncal,

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

while PDT provided significantly better outcomes in patients who are older with lesions on the lower extremities.[263] Safety results showed that while PDT causes moderate to severe pain during treatment administration, imiquimod and fluorouracil are more likely to cause moderate to severe local swelling, erosion, crust formation, itching, and wound infections.[231] Both cryotherapy and PDT are associated with pain during and after treatment, and data from a randomized trial indicate a trend toward a higher likelihood of pain with PDT.[259]

Nicotinamide in Reducing BCC Development

Data from phase II and phase III randomized trials indicated that treatment of actinic keratoses with nicotinamide reduced the occurrence of new BCCs, specifically by 20% at 12-month follow-up.[264,265] A systemic review and meta-analysis of randomized controlled trials determined that there was a significant association between nicotinamide and a reduction in BCCs and SCCs with a noted increased risk for digestive adverse events.[266] On the other hand, a phase III trial in recipients of organ transplants found no significant difference in the prevalence of SCC, BCC, or actinic keratosis between the groups randomly assigned to nicotinamide or placebo.[267] Other agents that might be effective for the prevention of BCC in individuals at high risk for developing NMSCs include celecoxib,[268] acitretin,[269] capecitabine,[270] and tazarotene.[271]

Systemic Therapy

For advanced BCC, systemic therapy is recommended as a treatment option for laBCC, mBCC, and nodal BCC after multidisciplinary consultation. Other options include surgery, RT, and palliation and best supportive care for certain patients. The systemic therapy options for BCC include hedgehog pathway inhibitor (HHI) and immunotherapy. Vismodegib and cemiplimab are currently recommended options for all advanced BCCs while sonidegib is only recommended for nodal BCC and laBCC.

Hedgehog Pathway Inhibitors

Vismodegib is an HHI approved by the U.S. Food and Drug Administration (FDA) for the treatment of adults with laBCC or mBCC that has recurred following surgery, or those who are not candidates for surgery or RT.[272] The 9-month follow-up data from the SHH4476g trial, a centrally reported, multicenter, phase I, open-label study, had an initial enrollment of 104 patients (laBCC N = 71, mBCC N = 33); however, pathology results excluded eight laBCC patients from the efficacy analysis (N = 63). This trial reported an overall response rate (ORR) of 30% in the mBCC group and 43% in the laBCC group, with a median duration of response (DOR) of 7.6 months and 9.5-month median progression-free survival (PFS).[273] A 39-month follow-up to these data from the ERIVANCE trial, an investigator-reported, multicenter, phase II trial, conveyed an ORR of 48.5% in the mBCC group and 60.3% in the laBCC group, with a median DOR of 14.8 months and 26.2 months for each group, respectively.[273-276] Results from these trials for vismodegib in BCC are summarized in Table 2. According to these data, nearly all patients treated with vismodegib experienced at least one treatment-emergent adverse event (TEAE), but a significant proportion of these were low grade (grade ≤2).[273-275] Serious adverse events (SAEs) occurred in 25% to 32% of patients in these studies. The most common adverse events (AEs) included muscle spasms, alopecia, taste loss, weight loss, decreased appetite, fatigue, nausea, and diarrhea.

For patients with RT-amenable unresectable laBCC, the combination of vismodegib with RT may be a valid approach. A phase II, multicenter, single-arm study evaluated this combination in 24 patients receiving 12 weeks on induction vismodegib followed by concurrent vismodegib with RT for 7 weeks where five patients were unable to complete induction.[277] At 1 year, locoregional control was achieved in 91% of patients with a 63% response rate after induction and 83% response rate after concurrent treatment. PFS and overall survival (OS) at 1 year were 100% and 96%,

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

respectively. At 5-year follow-up, PFS and OS rates dropped to 78% and 83%, respectively. Additionally, patients reported significant improvements in their quality of life.

Vismodegib has also been tested as BCC treatment and prophylaxis in patients with nevoid BCC syndrome. A randomized phase II study in patients with nevoid BCC syndrome and at least 10 operable BCC lesions found that vismodegib significantly reduced incidence of new BCC lesions compared with placebo, and also significantly reduced the size of existing lesions and the number of surgeries needed to remove BCC lesions.[278-280]

Sonidegib is another HHI FDA-approved agent for the treatment of patients with laBCC that has recurred following surgery or RT, or who are not candidates for surgery or RT.[272] Sonidegib is FDA approved for laBCC. The 42-month follow-up data from the centrally reported randomized, multicenter, phase II BOLT trial reported similar ORRs for the 200-mg and 800-mg doses tested among patients with laBCC (56% and 46%, respectively), while there was a 2-fold difference for patients with mBCC (8% and 17%, respectively).[281-285] This trial also reported, for each dose and patient group, median DOR and PFS results that are summarized in Table 2. The 30-month investigator-reviewed data for the BOLT trial analyzing only the 200-mg dose showed a higher ORR of 71.2% for laBCC and 23.1% for mBCC (Table 2).[283,286] As with vismodegib, nearly all patients experienced at least one AE, and the most common AEs were muscle spasms, dysgeusia, alopecia, nausea, weight decrease, and fatigue. Elevated creatinine kinase was also frequently observed and was one of the most common grade 3–4 AEs, along with elevated lipase.

A key limitation to HHI therapies is that advanced BCC can develop resistance, which limits DOR. A small investigator-initiated trial in patients with vismodegib-resistant advanced BCC observed no responses during treatment with sonidegib for a median of 6 weeks (range, 3–58 weeks), and in 5 of 9 patients with disease progression.[287]

Ongoing clinical research is exploring various dosing regimens of vismodegib and sonidegib in a variety of BCC treatment settings, including in the neoadjuvant setting, in patients with multiple BCCs or with radiation-induced multiple BCCs of the scalp, and as maintenance therapy after laBCC complete remission.[288-293]

Notably, in the neoadjuvant setting, while one trial reported negative results (unmet predefined complete histologic clearance rate),[289] results from two studies indicated vismodegib may reduce surgical defect area and allow for downstaging of the surgical procedure for laBCCs in functionally sensitive locations.[288,291] Vismodegib neoadjuvant efficacy and safety was evaluated in the multicenter, open-label, phase II VISMONEO study, 55 patients were enrolled and 80% (N = 44) presented with downstaging after a mean 6-month treatment duration.[291] This study reported an ORR of 71%, with 36% recurrence at the 3-year follow-up.[291] Additionally, a new surgical classification was presented in this publication according to morbidity where six predetermined stages (stage A–F) were used to determine the surgical procedure complexity with details for each area of the face.[291] Some of these studies included small numbers of patients, and thus their results need to be carefully interpreted.

Other HHIs are also being tested in patients with BCC to see if they can provide higher rates of response, more durable responses, responses in less advanced BCC, or responses in BCC resistant to vismodegib. Results from phase I–II trials with small BCC sample sizes (N < 40) have shown that itraconazole and saridegib can elicit responses in patients with BCC, although not in patients who previously received vismodegib.[294,295]

Immunotherapy

Cemiplimab-rwlc is an anti-programmed cell death protein 1 (PD-1) immunotherapy FDA-approved for patients with laBCC or mBCC previously treated with an HHI or for whom an HHI is not appropriate.[272] Cemiplimab is a recommended treatment option for certain patients with

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

advanced BCC including in the neoadjuvant setting for laBCC. A centrally reported, multicenter, phase II, open-label trial tested cemiplimab-rwlc (N = 84) for patients with laBCC where local invasion precluded complete resection or in locations for which surgery may have resulted in severe disfigurement or dysfunction or whose disease has progressed on or was intolerant to prior HHI therapy.[296,297] This study reported a median follow-up of 15 months, ORR of 31%, and grade 3–4 TEAEs in 48% of patients, while SAEs occurred in 35% of patients.[296] An analysis of patients with mBCC (N = 54) with an 8-month median follow-up resulted in 22% ORR per independent central review (ICR), with 2 complete responses and 10 partial responses.[298] Per ICR, there was a 3-month median time to response, 10-month median PFS, 50-month OS, and disease control rate of 63%. The most common TEAEs were fatigue and diarrhea.

Cytotoxic Chemotherapy

For patients who have progressed on or are ineligible for HHIs or cemiplimab, chemotherapy may be an alternative treatment option. The use of chemotherapeutic agents for BCCs has not been standardized and is limited to reviews and case reports.[299-305] Therefore, chemotherapy is not currently included as an NCCN-recommended treatment option. However, previous reports indicated that carboplatin/paclitaxel,[299] cisplatin/paclitaxel,[302] and doxorubicin[305] chemotherapy regimens yielded favorable responses in some patients with mBCC that had progressed on other treatments.

Follow-up

Follow-up for BCC should include a history and physical examination, along with a complete skin examination every 6 to 12 months for the first 5 years, and then at least annually for life. Imaging may be considered if clinical examination is insufficient for following the disease. Follow-up with a dermatologist is strongly recommended if any of the following criteria are met: past or imminent solid organ, marrow, or hematopoietic cell transplant; ≥1 cutaneous melanomas in the past 5 years; or ≥4 NMSCs within the past 5 years.

Imaging modality and targeted area should be at the discretion of the treating team based on the suspected extent of disease (ie, local, regional, metastatic). Histologic confirmation is sufficient to diagnose local recurrence, but imaging can be considered to assess extent of disease. As part of follow-up, the patients should be educated on sun protection and self-examination. For local recurrence, the primary treatment pathway for high-risk BCC should be followed. For locally advanced, nodal, and distant metastases, the appropriate path should be followed as found within *Advanced BCC* in the algorithm with consideration for multidisciplinary discussion and multimodality treatments.

An estimated 30% to 50% of patients with BCC will develop another BCC within 5 years.[130,133,306-309] This represents a 10-fold increase in risk compared to the general population.[307] Patients with a prior BCC are also at increased risk of developing SCC and cutaneous melanoma.[130,309] A prospective population-based cohort study found that development of a second BCC is most likely during the short-term follow-up period after diagnosis of the first lesion.[137] Therefore, close follow-up of patients with BCC in both the short- and long-term is critical.

Version 2.2026 © 2026 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

Table 1. Studies Comparing Superficial Therapies in Patients with Superficial BCC

Study	*Histologic Subtype*	*Tumor Locations*	*Treatments (n)*	*Efficacy*	*Cosmetic Outcome*
Phase III randomized trial Wang 2001[259]	Superficial and nodular	Trunk, limb, head, neck	Cryosurgery (39) ALA-PDT (44)	1-year recurrence: 15%, 25% } NS	Excellent: 8%, 50% } $P < .001$
Randomized trial Basset-Seguin 2008[260]	Superficial	Trunk, limb, head, neck, face	Cryotherapy (58) MAL-PDT (60)	5-year recurrence: 20%, 22% } NS	Excellent: 16%, 60% } $P = .00078$
Meta-analysis Roozeboom 2012[262]	Superficial	Locations depend on individual studies	Imiquimod (1088) PDT (934)	1-year tumor-free survival: 87%, 84% } NS	NR
Randomized, single-blind, non-inferiority trial Jansen 2018[215]	Superficial	Trunk, limb, head, neck	MAL-PDT (202) Imiquimod cream (198) Fluorouracil cream (201)	Treatment success[a]: 63%, 81% } $P < .001$; 81%, 70 } $P = .04$; } NS	Good/ excellent: 62%, 61%, 58% All comparisons NS

ALA, aminolevulinic acid; BCC, basal cell carcinoma; MAL, methyl aminolevulinate; NR, not reported; NS, no statistically significant difference; PDT, photodynamic therapy.
[a]Treatment success was defined as the product of the percentage of patients with clearance at 3 months by the percentage with sustained clearance during the next 9 months.

Version 2.2026 © 2026 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

Table 2. Hedgehog Pathway Inhibitors in Advanced BCC[a]

Study		*Tx[b]*	*Patients, n*		*Overall Response Rate[d]*		*Duration of Response, Median[c]*		*Progression-free Survival, Median[c]*		*Overall Survival, Median[c]*	
Name and References	*Design*		*laBCC*	*mBCC*	*laBCC*	*mBCC*	*laBCC*	*mBCC*	*laBCC*	*mBCC*	*laBCC*	*mBCC*
BOLT – 42-month follow-up NCT01327053[281,282,284,285]	Phase II RDB, CR	Soni 200 mg	66	13	56%	8%	26.1	24.0	22.1	13.1	NR	NR
		Soni 800 mg	128	23	46%	17%	23.3	NE	24.9	11.1	NR	NR
BOLT – 30-month follow-up NCT01327053[283,286]	Phase II RBD, IR	Soni 200 mg	66	13	71%	23.1%	15.7	17.7–18.4	NR	NR	NR	NR
SHH4476g – 9-month follow-up NCT00833417[273]	Phase I OL, CR	Vismo	63	33	43%	30%	7.6	7.6	9.5	9.5	NR	NR
ERIVANCE – 39-month follow-up NCT00833417[274-276]	Phase II OL, IR	Vismo	63	33	60%	49%	26.2	14.8	12.9	9.3	NE	33.4

BCC, basal cell carcinoma; CR, centrally reviewed; IR, investigator reviewed; laBCC, locally advanced BCC; mBCC, metastatic BCC; NE, not reached; NR, not reported; OL, open-label; RDB, randomized double-blind; Soni, sonidegib; Tx, treatment; Vismo, vismodegib.
[a]Trials included patients with advanced BCC that was inappropriate for surgery or RT.
[b]Inhibitors were taken orally once daily. Vismodegib dose was 150 mg.
[c]Times are reported in months.
[d]Response criteria varied between studies.

Version 2.2026 © 2026 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

References

1. Asgari MM, Moffet HH, Ray GT, Quesenberry CP. Trends in basal cell carcinoma incidence and identification of high-risk subgroups, 1998-2012. JAMA Dermatol 2015;151:976–981. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26039887.

2. Siegel RL, Miller KD, Jemal A. Cancer statistics, 2016. CA Cancer J Clin 2016;66:7–30. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26742998.

3. Rogers HW, Weinstock MA, Feldman SR, Coldiron BM. Incidence estimate of nonmelanoma skin cancer (keratinocyte carcinomas) in the u.S. Population, 2012. JAMA Dermatol 2015;151:1081–1086. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25928283.

4. Abbas M, Kalia S. Trends in non-melanoma skin cancer (basal cell carcinoma and squamous cell carcinoma) in canada: A descriptive analysis of available data. J Cutan Med Surg 2016;20:166–175. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26458408.

5. Rudolph C, Schnoor M, Eisemann N, Katalinic A. Incidence trends of nonmelanoma skin cancer in Germany from 1998 to 2010. J Dtsch Dermatol Ges 2015;13:788–797. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26213814.

6. Muzic JG, Schmitt AR, Wright AC, et al. Incidence and trends of basal cell carcinoma and cutaneous squamous cell carcinoma: A population-based study in olmsted county, minnesota, 2000 to 2010. Mayo Clin Proc 2017;92:890–898. Available at: https://www.ncbi.nlm.nih.gov/pubmed/28522111.

7. Christenson LJ, Borrowman TA, Vachon CM, et al. Incidence of basal cell and squamous cell carcinomas in a population younger than 40 years. JAMA 2005;294:681–690. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16091570.

8. von Domarus H, Stevens PJ. Metastatic basal cell carcinoma. Report of five cases and review of 170 cases in the literature. J Am Acad Dermatol 1984;10:1043–1060. Available at: https://www.ncbi.nlm.nih.gov/pubmed/6736323.

9. Nguyen-Nielsen M, Wang L, Pedersen L, et al. The incidence of metastatic basal cell carcinoma (mBCC) in Denmark, 1997-2010. Eur J Dermatol 2015;25:463–468. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26105129.

10. Wadhera A, Fazio M, Bricca G, Stanton O. Metastatic basal cell carcinoma: a case report and literature review. How accurate is our incidence data? Dermatol Online J 2006;12:7. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16962022.

11. Kricker A, Armstrong BK, English DR, Heenan PJ. Does intermittent sun exposure cause basal cell carcinoma? a case-control study in Western Australia. Int J Cancer 1995;60:489–494. Available at: https://www.ncbi.nlm.nih.gov/pubmed/7829262.

12. Kricker A, Armstrong BK, English DR, Heenan PJ. A dose-response curve for sun exposure and basal cell carcinoma. Int J Cancer 1995;60:482–488. Available at: https://www.ncbi.nlm.nih.gov/pubmed/7829261.

13. Zanetti R, Rosso S, Martinez C, et al. Comparison of risk patterns in carcinoma and melanoma of the skin in men: a multi-centre case-case-control study. Br J Cancer 2006;94:743–751. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16495934.

14. Gallagher RP, Hill GB, Bajdik CD, et al. Sunlight exposure, pigmentary factors, and risk of nonmelanocytic skin cancer. I. Basal cell carcinoma. Arch Dermatol 1995;131:157–163. Available at: https://www.ncbi.nlm.nih.gov/pubmed/7857111.

15. Ramsay HM, Fryer AA, Hawley CM, et al. Factors associated with nonmelanoma skin cancer following renal transplantation in Queensland, Australia. J Am Acad Dermatol 2003;49:397–406. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12963901.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

16. Kaskel P, Lange U, Sander S, et al. Ultraviolet exposure and risk of melanoma and basal cell carcinoma in Ulm and Dresden, Germany. J Eur Acad Dermatol Venereol 2015;29:134–142. Available at: https://www.ncbi.nlm.nih.gov/pubmed/24684198.

17. Khalesi M, Whiteman DC, Tran B, et al. A meta-analysis of pigmentary characteristics, sun sensitivity, freckling and melanocytic nevi and risk of basal cell carcinoma of the skin. Cancer Epidemiol 2013;37:534–543. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23849507.

18. Walther U, Kron M, Sander S, et al. Risk and protective factors for sporadic basal cell carcinoma: results of a two-centre case-control study in southern Germany. Clinical actinic elastosis may be a protective factor. Br J Dermatol 2004;151:170–178. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15270887.

19. Box NF, Duffy DL, Irving RE, et al. Melanocortin-1 receptor genotype is a risk factor for basal and squamous cell carcinoma. J Invest Dermatol 2001;116:224–229. Available at: https://www.ncbi.nlm.nih.gov/pubmed/11179997.

20. Lock-Andersen J, Drzewiecki KT, Wulf HC. Eye and hair colour, skin type and constitutive skin pigmentation as risk factors for basal cell carcinoma and cutaneous malignant melanoma. A Danish case-control study. Acta Derm Venereol 1999;79:74–80. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10086866.

21. Chinem VP, Miot HA. Prevalence of actinic skin lesions in patients with basal cell carcinoma of the head: a case-control study. Rev Assoc Med Bras (1992) 2012;58:188–196. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22569613.

22. Wu S, Han J, Li WQ, et al. Basal-cell carcinoma incidence and associated risk factors in U.S. women and men. Am J Epidemiol 2013;178:890–897. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23828250.

23. Perkins JL, Liu Y, Mitby PA, et al. Nonmelanoma skin cancer in survivors of childhood and adolescent cancer: a report from the childhood cancer survivor study. J Clin Oncol 2005;23:3733–3741. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15923570.

24. Karagas MR, Nelson HH, Zens MS, et al. Squamous cell and basal cell carcinoma of the skin in relation to radiation therapy and potential modification of risk by sun exposure. Epidemiology 2007;18:776–784. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17917604.

25. Watt TC, Inskip PD, Stratton K, et al. Radiation-related risk of basal cell carcinoma: a report from the Childhood Cancer Survivor Study. J Natl Cancer Inst 2012;104:1240–1250. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22835387.

26. Schwartz JL, Kopecky KJ, Mathes RW, et al. Basal cell skin cancer after total-body irradiation and hematopoietic cell transplantation. Radiat Res 2009;171:155–163. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19267540.

27. Karagas MR, McDonald JA, Greenberg ER, et al. Risk of basal cell and squamous cell skin cancers after ionizing radiation therapy. For The Skin Cancer Prevention Study Group. J Natl Cancer Inst 1996;88:1848–1853. Available at: https://www.ncbi.nlm.nih.gov/pubmed/8961975.

28. PubMed Overview. Available at: https://pubmed.ncbi.nlm.nih.gov/about/.

29. Freedman-Cass DA, Fischer T, Alpert AB, et al. The value and process of inclusion: Using sensitive, respectful, and inclusive language and images in nccn content. J Natl Compr Canc Netw 2023;21:434–441. Available at: https://www.ncbi.nlm.nih.gov/pubmed/37156485.

30. Lesiak A, Sobolewska-Sztychny D, Majak P, et al. Relation between sonic hedgehog pathway gene polymorphisms and basal cell carcinoma development in the Polish population. Arch Dermatol Res 2016;308:39–47. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26590974.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

31. Reifenberger J, Wolter M, Weber RG, et al. Missense mutations in SMOH in sporadic basal cell carcinomas of the skin and primitive neuroectodermal tumors of the central nervous system. Cancer Res 1998;58:1798–1803. Available at: https://www.ncbi.nlm.nih.gov/pubmed/9581815.

32. Xie J, Murone M, Luoh SM, et al. Activating Smoothened mutations in sporadic basal-cell carcinoma. Nature 1998;391:90–92. Available at: https://www.ncbi.nlm.nih.gov/pubmed/9422511.

33. Gailani MR, Bale SJ, Leffell DJ, et al. Developmental defects in Gorlin syndrome related to a putative tumor suppressor gene on chromosome 9. Cell 1992;69:111–117. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1348213.

34. Soufir N, Gerard B, Portela M, et al. PTCH mutations and deletions in patients with typical nevoid basal cell carcinoma syndrome and in patients with a suspected genetic predisposition to basal cell carcinoma: a French study. Br J Cancer 2006;95:548–553. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16909134.

35. Ling G, Ahmadian A, Persson A, et al. PATCHED and p53 gene alterations in sporadic and hereditary basal cell cancer. Oncogene 2001;20:7770–7778. Available at: https://www.ncbi.nlm.nih.gov/pubmed/11753655.

36. Heitzer E, Lassacher A, Quehenberger F, et al. UV fingerprints predominate in the PTCH mutation spectra of basal cell carcinomas independent of clinical phenotype. J Invest Dermatol 2007;127:2872–2881. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17597822.

37. Danaee H, Karagas MR, Kelsey KT, et al. Allelic loss at Drosophila patched gene is highly prevalent in Basal and squamous cell carcinomas of the skin. J Invest Dermatol 2006;126:1152–1158. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16484983.

38. Reifenberger J, Wolter M, Knobbe CB, et al. Somatic mutations in the PTCH, SMOH, SUFUH and TP53 genes in sporadic basal cell carcinomas. Br J Dermatol 2005;152:43–51. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15656799.

39. Kim MY, Park HJ, Baek SC, et al. Mutations of the p53 and PTCH gene in basal cell carcinomas: UV mutation signature and strand bias. J Dermatol Sci 2002;29:1–9. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12007715.

40. Gailani MR, Stahle-Backdahl M, Leffell DJ, et al. The role of the human homologue of Drosophila patched in sporadic basal cell carcinomas. Nat Genet 1996;14:78–81. Available at: https://www.ncbi.nlm.nih.gov/pubmed/8782823.

41. Zhang H, Ping XL, Lee PK, et al. Role of PTCH and p53 genes in early-onset basal cell carcinoma. Am J Pathol 2001;158:381–385. Available at: https://www.ncbi.nlm.nih.gov/pubmed/11159175.

42. Ziegler A, Leffell DJ, Kunala S, et al. Mutation hotspots due to sunlight in the p53 gene of nonmelanoma skin cancers. Proc Natl Acad Sci U S A 1993;90:4216–4220. Available at: https://www.ncbi.nlm.nih.gov/pubmed/8483937.

43. Oluwasanmi JO, Williams AO, Alli AF. Superficial cancer in Nigeria. Br J Cancer 1969;23:714–728. Available at: https://www.ncbi.nlm.nih.gov/pubmed/5367332.

44. Yakubu A, Mabogunje OA. Skin cancer in Zaria, Nigeria. Trop Doct 1995;25 Suppl 1:63–67. Available at: https://www.ncbi.nlm.nih.gov/pubmed/7879275.

45. Kraemer KH, Lee MM, Scotto J. Xeroderma pigmentosum. Cutaneous, ocular, and neurologic abnormalities in 830 published cases. Arch Dermatol 1987;123:241–250. Available at: https://www.ncbi.nlm.nih.gov/pubmed/3545087.

46. Bradford PT, Goldstein AM, Tamura D, et al. Cancer and neurologic degeneration in xeroderma pigmentosum: long term follow-up characterises the role of DNA repair. J Med Genet 2011;48:168–176. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21097776.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

47. Kraemer KH, Lee MM, Andrews AD, Lambert WC. The role of sunlight and DNA repair in melanoma and nonmelanoma skin cancer. The xeroderma pigmentosum paradigm. Arch Dermatol 1994;130:1018–1021. Available at: https://www.ncbi.nlm.nih.gov/pubmed/8053698.

48. Kraemer KH, Lee MM, Scotto J. DNA repair protects against cutaneous and internal neoplasia: evidence from xeroderma pigmentosum. Carcinogenesis 1984;5:511–514. Available at: https://www.ncbi.nlm.nih.gov/pubmed/6705149.

49. Couve-Privat S, Le Bret M, Traiffort E, et al. Functional analysis of novel sonic hedgehog gene mutations identified in basal cell carcinomas from xeroderma pigmentosum patients. Cancer Res 2004;64:3559–3565. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15150112.

50. Couve-Privat S, Bouadjar B, Avril MF, et al. Significantly high levels of ultraviolet-specific mutations in the smoothened gene in basal cell carcinomas from DNA repair-deficient xeroderma pigmentosum patients. Cancer Res 2002;62:7186–7189. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12499255.

51. Miller KL, Karagas MR, Kraft P, et al. XPA, haplotypes, and risk of basal and squamous cell carcinoma. Carcinogenesis 2006;27:1670–1675. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16513681.

52. Clements S, Khachemoune A. Upstaging of basal cell and squamous cell carcinomas during definitive surgery: a review of predictive preoperative clinical and histologic features. Arch Dermatol Res 2021;313:319–325. Available at: https://www.ncbi.nlm.nih.gov/pubmed/33108525.

53. Singh B, Dorelles A, Konnikov N, Nguyen BM. Detection of high-risk histologic features and tumor upstaging of nonmelanoma skin cancers on debulk analysis: A quantitative systematic review. Dermatol Surg 2017;43:1003–1011. Available at: https://www.ncbi.nlm.nih.gov/pubmed/28654579.

54. Chen J, Ruczinski I, Jorgensen TJ, et al. Nonmelanoma skin cancer and risk for subsequent malignancy. J Natl Cancer Inst 2008;100:1215–1222. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18728282.

55. Morgan FC, Ruiz ES, Karia PS, et al. Brigham and Women's Hospital tumor classification system for basal cell carcinoma identifies patients with risk of metastasis and death. J Am Acad Dermatol 2021;85:582–587. Available at: https://www.ncbi.nlm.nih.gov/pubmed/33497751.

56. Gandhi MR, Panizza B, Kennedy D. Detecting and defining the anatomic extent of large nerve perineural spread of malignancy: comparing "targeted" MRI with the histologic findings following surgery. Head Neck 2011;33:469–475. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20645285.

57. Williams LS, Mancuso AA, Mendenhall WM. Perineural spread of cutaneous squamous and basal cell carcinoma: CT and MR detection and its impact on patient management and prognosis. Int J Radiat Oncol Biol Phys 2001;49:1061–1069. Available at: https://www.ncbi.nlm.nih.gov/pubmed/11240248.

58. Boeta-Angeles L, Bennett RG. Features associated with recurrence (basal cell carcinoma). In: Miller SJ, Maloney ME, eds. Cutaneous Oncology Pathophysiology, diagnosis, and management. Malden, MA: Blackwell Science; 1998:646–656.

59. Silverman MK, Kopf AW, Bart RS, et al. Recurrence rates of treated basal cell carcinomas. Part 3: Surgical excision. J Dermatol Surg Oncol 1992;18:471–476. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1592998.

60. Silverman MK, Kopf AW, Grin CM, et al. Recurrence rates of treated basal cell carcinomas. Part 2: Curettage-electrodesiccation. J Dermatol Surg Oncol 1991;17:720–726. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1820764.

61. Dubin N, Kopf AW. Multivariate risk score for recurrence of cutaneous basal cell carcinomas. Arch Dermatol 1983;119:373–377. Available at: https://www.ncbi.nlm.nih.gov/pubmed/6847215.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

62. Bogelund FS, Philipsen PA, Gniadecki R. Factors affecting the recurrence rate of basal cell carcinoma. Acta Derm Venereol 2007;87:330–334. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17598036.

63. Rigel DS, Robins P, Friedman RJ. Predicting recurrence of basal-cell carcinomas treated by microscopically controlled excision: a recurrence index score. J Dermatol Surg Oncol 1981;7:807–810. Available at: https://www.ncbi.nlm.nih.gov/pubmed/7298981.

64. Spiller WF, Spiller RF. Treatment of basal cell epithelioma by curettage and electrodesiccation. J Am Acad Dermatol 1984;11:808–814. Available at: https://www.ncbi.nlm.nih.gov/pubmed/6512037.

65. van Iersel CA, van de Velden HV, Kusters CD, et al. Prognostic factors for a subsequent basal cell carcinoma: implications for follow-up. Br J Dermatol 2005;153:1078–1080. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16225637.

66. Petrovich Z, Kuisk H, Langholz B, et al. Treatment results and patterns of failure in 646 patients with carcinoma of the eyelids, pinna, and nose. Am J Surg 1987;154:447–450. Available at: https://www.ncbi.nlm.nih.gov/pubmed/3661851.

67. Swanson NA. Mohs surgery. Technique, indications, applications, and the future. Arch Dermatol 1983;119:761–773. Available at: https://www.ncbi.nlm.nih.gov/pubmed/6351758.

68. Yalcin O, Sezer E, Kabukcuoglu F, et al. Presence of ulceration, but not high risk zone location, correlates with unfavorable histopathological subtype in facial basal cell carcinoma. Int J Clin Exp Pathol 2015;8:15448–15453. Available at: http://www.ncbi.nlm.nih.gov/pubmed/26823913.

69. van Loo E, Mosterd K, Krekels GA, et al. Surgical excision versus Mohs' micrographic surgery for basal cell carcinoma of the face: A randomised clinical trial with 10 year follow-up. Eur J Cancer 2014;50:3011–3020. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25262378.

70. Venturi F, Erbacci V, Veronesi G, et al. Basosquamous carcinoma: Comprehensive epidemiological, clinical, dermoscopic, and confocal features from a single center institution. Skin Res Technol 2024;30:e70012. Available at: https://www.ncbi.nlm.nih.gov/pubmed/39137046.

71. Silverman MK, Kopf AW, Grin CM, et al. Recurrence rates of treated basal cell carcinomas. Part 1: Overview. J Dermatol Surg Oncol 1991;17:713–718. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1890243.

72. American Academy of Dermatology, American College of Mohs Surgery, American Society for Dermatologic Surgery Association, et al. AAD/ACMS/ASDSA/ASMS 2012 appropriate use criteria for Mohs micrographic surgery: a report of the American Academy of Dermatology, American College of Mohs Surgery, American Society for Dermatologic Surgery Association, and the American Society for Mohs Surgery. Dermatol Surg 2012;38:1582–1603. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22958088.

73. Dixon AY, Lee SH, McGregor DH. Histologic features predictive of basal cell carcinoma recurrence: results of a multivariate analysis. J Cutan Pathol 1993;20:137–142. Available at: https://www.ncbi.nlm.nih.gov/pubmed/8320358.

74. Jacobs GH, Rippey JJ, Altini M. Prediction of aggressive behavior in basal cell carcinoma. Cancer 1982;49:533–537. Available at: https://www.ncbi.nlm.nih.gov/pubmed/7059912.

75. de Rosa G, Vetrani A, Zeppa P, et al. Comparative morphometric analysis of aggressive and ordinary basal cell carcinoma of the skin. Cancer 1990;65:544–549. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2297645.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

76. Codazzi D, Van Der Velden J, Carminati M, et al. Positive compared with negative margins in a single-centre retrospective study on 3957 consecutive excisions of basal cell carcinomas. Associated risk factors and preferred surgical management. J Plast Surg Hand Surg 2014;48:38–43. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23731130.

77. Krynitz B, Olsson H, Lundh Rozell B, et al. Risk of basal cell carcinoma in Swedish organ transplant recipients: a population-based study. Br J Dermatol 2016;174:95–103. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26333521.

78. Mackintosh LJ, Geddes CC, Herd RM. Skin tumours in the West of Scotland renal transplant population. Br J Dermatol 2013;168:1047–1053. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23137036.

79. Bernat Garcia J, Morales Suarez-Varela M, Vilata JJ, et al. Risk factors for non-melanoma skin cancer in kidney transplant patients in a Spanish population in the Mediterranean region. Acta Derm Venereol 2013;93:422–427. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23303600.

80. Karczewski M, Stronka M, Karczewski J, Wiktorowicz K. Skin cancer following kidney transplantation: a single-center experience. Transplant Proc 2011;43:3760–3761. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22172842.

81. Bordea C, Wojnarowska F, Millard PR, et al. Skin cancers in renal-transplant recipients occur more frequently than previously recognized in a temperate climate. Transplantation 2004;77:574–579. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15084938.

82. DePry JL, Vyas R, Lazarus HM, et al. Cutaneous malignant neoplasms in hematopoietic cell transplant recipients: A systematic review. JAMA Dermatol 2015;151:775–782. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25902409.

83. Stern RS, Liebman EJ, Vakeva L. Oral psoralen and ultraviolet-A light (PUVA) treatment of psoriasis and persistent risk of nonmelanoma skin cancer. PUVA Follow-up Study. J Natl Cancer Inst 1998;90:1278–1284. Available at: https://www.ncbi.nlm.nih.gov/pubmed/9731734.

84. Archier E, Devaux S, Castela E, et al. Carcinogenic risks of psoralen UV-A therapy and narrowband UV-B therapy in chronic plaque psoriasis: a systematic literature review. J Eur Acad Dermatol Venereol 2012;26 Suppl 3:22–31. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22512677.

85. Park GH, Chang SE, Won CH, et al. Incidence of primary skin cancer after organ transplantation: An 18-year single-center experience in Korea. J Am Acad Dermatol 2014;70:465–472. Available at: https://www.ncbi.nlm.nih.gov/pubmed/24342756.

86. Jensen AO, Svaerke C, Farkas D, et al. Skin cancer risk among solid organ recipients: a nationwide cohort study in Denmark. Acta Derm Venereol 2010;90:474–479. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20814621.

87. Hartevelt MM, Bavinck JN, Kootte AM, et al. Incidence of skin cancer after renal transplantation in The Netherlands. Transplantation 1990;49:506–509. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2316011.

88. Harwood CA, Mesher D, McGregor JM, et al. A surveillance model for skin cancer in organ transplant recipients: a 22-year prospective study in an ethnically diverse population. Am J Transplant 2013;13:119–129. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23072567.

89. Brewer JD, Colegio OR, Phillips PK, et al. Incidence of and risk factors for skin cancer after heart transplant. Arch Dermatol 2009;145:1391–1396. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20026847.

90. Rashtak S, Dierkhising RA, Kremers WK, et al. Incidence and risk factors for skin cancer following lung transplantation. J Am Acad Dermatol 2015;72:92–98. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25440431.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

91. Fortina AB, Piaserico S, Caforio AL, et al. Immunosuppressive level and other risk factors for basal cell carcinoma and squamous cell carcinoma in heart transplant recipients. Arch Dermatol 2004;140:1079–1085. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15381547.

92. Kanitakis J, Alhaj-Ibrahim L, Euvrard S, Claudy A. Basal cell carcinomas developing in solid organ transplant recipients: clinicopathologic study of 176 cases. Arch Dermatol 2003;139:1133–1137. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12975154.

93. Harwood CA, Proby CM, McGregor JM, et al. Clinicopathologic features of skin cancer in organ transplant recipients: a retrospective case-control series. J Am Acad Dermatol 2006;54:290–300. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16443060.

94. Lott DG, Manz R, Koch C, Lorenz RR. Aggressive behavior of nonmelanotic skin cancers in solid organ transplant recipients. Transplantation 2010;90:683–687. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20808266.

95. Martin H, Strong E, Spiro RH. Radiation-induced skin cancer of the head and neck. Cancer 1970;25:61–71. Available at: https://www.ncbi.nlm.nih.gov/pubmed/4312028.

96. Lichter MD, Karagas MR, Mott LA, et al. Therapeutic ionizing radiation and the incidence of basal cell carcinoma and squamous cell carcinoma. The New Hampshire Skin Cancer Study Group. Arch Dermatol 2000;136:1007–1011. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10926736.

97. Dixon AY, Lee SH, McGregor DH. Factors predictive of recurrence of basal cell carcinoma. Am J Dermatopathol 1989;11:222–232. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2729527.

98. Smeets NW, Kuijpers DI, Nelemans P, et al. Mohs' micrographic surgery for treatment of basal cell carcinoma of the face--results of a retrospective study and review of the literature. Br J Dermatol 2004;151:141–147. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15270883.

99. Cigna E, Tarallo M, Maruccia M, et al. Basal cell carcinoma: 10 years of experience. J Skin Cancer 2011;2011:476362. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21151696.

100. Szewczyk MP, Pazdrowski J, Danczak-Pazdrowska A, et al. Analysis of selected recurrence risk factors after treatment of head and neck basal cell carcinoma. Postepy Dermatol Alergol 2014;31:146–151. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25097485.

101. Bartos V, Pokorny D, Zacharova O, et al. Recurrent basal cell carcinoma: a clinicopathological study and evaluation of histomorphological findings in primary and recurrent lesions. Acta Dermatovenerol Alp Pannonica Adriat 2011;20:67–75. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21993704.

102. Sartore L, Lancerotto L, Salmaso M, et al. Facial basal cell carcinoma: analysis of recurrence and follow-up strategies. Oncol Rep 2011;26:1423–1429. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21922143.

103. Sloane JP. The value of typing basal cell carcinomas in predicting recurrence after surgical excision. Br J Dermatol 1977;96:127–132. Available at: https://www.ncbi.nlm.nih.gov/pubmed/843446.

104. Costantino D, Lowe L, Brown DL. Basosquamous carcinoma-an under-recognized, high-risk cutaneous neoplasm: case study and review of the literature. J Plast Reconstr Aesthet Surg 2006;59:424–428. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16756261.

105. Martin RC, 2nd, Edwards MJ, Cawte TG, et al. Basosquamous carcinoma: analysis of prognostic factors influencing recurrence. Cancer 2000;88:1365–1369. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10717618.

106. Garcia C, Poletti E, Crowson AN. Basosquamous carcinoma. J Am Acad Dermatol 2009;60:137–143. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19103364.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

107. Wermker K, Roknic N, Goessling K, et al. Basosquamous carcinoma of the head and neck: clinical and histologic characteristics and their impact on disease progression. Neoplasia 2015;17:301–305. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25810014.

108. Hassanein AM, Proper SA, Depcik-Smith ND, Flowers FP. Peritumoral fibrosis in basal cell and squamous cell carcinoma mimicking perineural invasion: potential pitfall in Mohs micrographic surgery. Dermatol Surg 2005;31:1101–1106. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16164857.

109. Lin C, Tripcony L, Keller J, et al. Perineural infiltration of cutaneous squamous cell carcinoma and basal cell carcinoma without clinical features. Int J Radiat Oncol Biol Phys 2012;82:334–340. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21093171.

110. Garcia-Serra A, Hinerman RW, Mendenhall WM, et al. Carcinoma of the skin with perineural invasion. Head Neck 2003;25:1027–1033. Available at: https://www.ncbi.nlm.nih.gov/pubmed/14648861.

111. Lin C, Tripcony L, Keller J, et al. Cutaneous carcinoma of the head and neck with clinical features of perineural infiltration treated with radiotherapy. Clin Oncol (R Coll Radiol) 2013;25:362–367. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23489870.

112. Leibovitch I, Huilgol SC, Selva D, et al. Basal cell carcinoma treated with Mohs surgery in Australia III. Perineural invasion. J Am Acad Dermatol 2005;53:458–463. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16112353.

113. Jackson JE, Dickie GJ, Wiltshire KL, et al. Radiotherapy for perineural invasion in cutaneous head and neck carcinomas: toward a risk-adapted treatment approach. Head Neck 2009;31:604–610. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19132719.

114. Ratner D, Lowe L, Johnson TM, Fader DJ. Perineural spread of basal cell carcinomas treated with Mohs micrographic surgery. Cancer 2000;88:1605–1613. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10738219.

115. Brown CI, Perry AE. Incidence of perineural invasion in histologically aggressive types of basal cell carcinoma. Am J Dermatopathol 2000;22:123–125. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10770431.

116. Galloway TJ, Morris CG, Mancuso AA, et al. Impact of radiographic findings on prognosis for skin carcinoma with clinical perineural invasion. Cancer 2005;103:1254–1257. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15693020.

117. Balamucki CJ, DeJesus R, Galloway TJ, et al. Impact of radiographic findings on for prognosis skin cancer with perineural invasion. Am J Clin Oncol 2015;38:248–251. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23648439.

118. Cernea CR, Ferraz AR, de Castro IV, et al. Perineural invasion in aggressive skin carcinomas of the head and neck. Potentially dangerous but frequently overlooked. ORL J Otorhinolaryngol Relat Spec 2009;71:21–26. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18946230.

119. Leffell DJ, Headington JT, Wong DS, Swanson NA. Aggressive-growth basal cell carcinoma in young adults. Arch Dermatol 1991;127:1663–1667. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1952969.

120. McCormack CJ, Kelly JW, Dorevitch AP. Differences in age and body site distribution of the histological subtypes of basal cell carcinoma. A possible indicator of differing causes. Arch Dermatol 1997;133:593–596. Available at: https://www.ncbi.nlm.nih.gov/pubmed/9158412.

121. Bastiaens MT, Hoefnagel JJ, Bruijn JA, et al. Differences in age, site distribution, and sex between nodular and superficial basal cell carcinoma indicate different types of tumors. J Invest Dermatol 1998;110:880–884. Available at: https://www.ncbi.nlm.nih.gov/pubmed/9620293.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

122. Scrivener Y, Grosshans E, Cribier B. Variations of basal cell carcinomas according to gender, age, location and histopathological subtype. Br J Dermatol 2002;147:41–47. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12100183.

123. Raasch BA, Buettner PG, Garbe C. Basal cell carcinoma: histological classification and body-site distribution. Br J Dermatol 2006;155:401–407. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16882181.

124. Dinehart SM, Dodge R, Stanley WE, et al. Basal cell carcinoma treated with Mohs surgery. A comparison of 54 younger patients with 1050 older patients. J Dermatol Surg Oncol 1992;18:560–566. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1624629.

125. Milroy CJ, Horlock N, Wilson GD, Sanders R. Aggressive basal cell carcinoma in young patients: fact or fiction? Br J Plast Surg 2000;53:393–396. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10876275.

126. Roudier-Pujol C, Auperin A, Nguyen T, et al. Basal cell carcinoma in young adults: not more aggressive than in older patients. Dermatology 1999;199:119–123. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10559576.

127. Lear JT, Smith AG, Bowers B, et al. Truncal tumor site is associated with high risk of multiple basal cell carcinoma and is influenced by glutathione S-transferase, GSTT1, and cytochrome P450, CYP1A1 genotypes, and their interaction. J Invest Dermatol 1997;108:519–522. Available at: https://www.ncbi.nlm.nih.gov/pubmed/9077484.

128. Ramachandran S, Fryer AA, Lovatt T, et al. The rate of increase in the numbers of primary sporadic basal cell carcinomas during follow up is associated with age at first presentation. Carcinogenesis 2002;23:2051–2054. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12507928.

129. Cheretis C, Angelidou E, Dietrich F, et al. Prognostic value of computer-assisted morphological and morphometrical analysis for detecting the recurrence tendency of basal cell carcinoma. Med Sci Monit 2008;14:MT13–19. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18443558.

130. Karagas MR, Stukel TA, Greenberg ER, et al. Risk of subsequent basal cell carcinoma and squamous cell carcinoma of the skin among patients with prior skin cancer. Skin Cancer Prevention Study Group. JAMA 1992;267:3305–3310. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1597912.

131. Lovatt TJ, Lear JT, Bastrilles J, et al. Associations between ultraviolet radiation, basal cell carcinoma site and histology, host characteristics, and rate of development of further tumors. J Am Acad Dermatol 2005;52:468–473. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15761425.

132. Richmond-Sinclair NM, Pandeya N, Williams GM, et al. Clinical signs of photodamage are associated with basal cell carcinoma multiplicity and site: a 16-year longitudinal study. Int J Cancer 2010;127:2622–2629. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20196068.

133. Flohil SC, Koljenovic S, de Haas ER, et al. Cumulative risks and rates of subsequent basal cell carcinomas in the Netherlands. Br J Dermatol 2011;165:874–881. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21711333.

134. Verkouteren JAC, Smedinga H, Steyerberg EW, et al. Predicting the Risk of a Second Basal Cell Carcinoma. J Invest Dermatol 2015;135:2649–2656. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26121210.

135. Milan T, Pukkala E, Verkasalo PK, et al. Subsequent primary cancers after basal-cell carcinoma: A nationwide study in Finland from 1953 to 1995. Int J Cancer 2000;87:283–288. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10861488.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

136. McCusker M, Basset-Seguin N, Dummer R, et al. Metastatic basal cell carcinoma: prognosis dependent on anatomic site and spread of disease. Eur J Cancer 2014;50:774–783. Available at: https://www.ncbi.nlm.nih.gov/pubmed/24412051.

137. Kiiski V, de Vries E, Flohil SC, et al. Risk factors for single and multiple basal cell carcinomas. Arch Dermatol 2010;146:848–855. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20713815.

138. Rowe DE, Carroll RJ, Day CL, Jr. Long-term recurrence rates in previously untreated (primary) basal cell carcinoma: implications for patient follow-up. J Dermatol Surg Oncol 1989;15:315–328. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2646336.

139. Barlow JO, Zalla MJ, Kyle A, et al. Treatment of basal cell carcinoma with curettage alone. J Am Acad Dermatol 2006;54:1039–1045. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16713459.

140. Thissen MR, Neumann MH, Schouten LJ. A systematic review of treatment modalities for primary basal cell carcinomas. Arch Dermatol 1999;135:1177–1183. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10522664.

141. Chren MM, Torres JS, Stuart SE, et al. Recurrence after treatment of nonmelanoma skin cancer: a prospective cohort study. Arch Dermatol 2011;147:540–546. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21576572.

142. Julian C, Bowers PW, Pritchard C. A comparative study of the effects of disposable and Volkmann spoon curettes in the treatment of basal cell carcinoma. Br J Dermatol 2009;161:1407–1409. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19681879.

143. Blixt E, Nelsen D, Stratman E. Recurrence rates of aggressive histologic types of basal cell carcinoma after treatment with electrodesiccation and curettage alone. Dermatol Surg 2013;39:719–725. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23379543.

144. Rodriguez-Vigil T, Vazquez-Lopez F, Perez-Oliva N. Recurrence rates of primary basal cell carcinoma in facial risk areas treated with curettage and electrodesiccation. J Am Acad Dermatol 2007;56:91–95. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17190625.

145. Kopf AW, Bart RS, Schrager D, et al. Curettage-electrodesiccation treatment of basal cell carcinomas. Arch Dermatol 1977;113:439–443. Available at: https://www.ncbi.nlm.nih.gov/pubmed/848972.

146. Rowe DE, Carroll RJ, Day CL, Jr. Mohs surgery is the treatment of choice for recurrent (previously treated) basal cell carcinoma. J Dermatol Surg Oncol 1989;15:424–431. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2925988.

147. Emmett AJ, Broadbent GD. Shave excision of superficial solar skin lesions. Plast Reconstr Surg 1987;80:47–54. Available at: https://www.ncbi.nlm.nih.gov/pubmed/3602160.

148. Dando EE, Abban C, Shehu Wingrove A, et al. Deep shave removal of suspected basal cell carcinoma: A prospective study. Dermatol Surg 2023;49:130–134. Available at: https://www.ncbi.nlm.nih.gov/pubmed/36728062.

149. Abramson AK, Krasny MJ, Goldman GD. Tangential shave removal of basal cell carcinoma. Dermatol Surg 2013;39:387–392. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23279298.

150. Wu X, Elkin EB, Jason Chen CS, Marghoob A. Traditional versus streamlined management of basal cell carcinoma (BCC): A cost analysis. J Am Acad Dermatol 2015;73:791–798. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26341142.

151. Kuijpers DI, Thissen MR, Berretty PJ, et al. Surgical excision versus curettage plus cryosurgery in the treatment of basal cell carcinoma. Dermatol Surg 2007;33:579–587. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17451581.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

152. Rhodes LE, de Rie MA, Leifsdottir R, et al. Five-year follow-up of a randomized, prospective trial of topical methyl aminolevulinate photodynamic therapy vs surgery for nodular basal cell carcinoma. Arch Dermatol 2007;143:1131–1136. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17875873.

153. Wetzig T, Woitek M, Eichhorn K, et al. Surgical excision of basal cell carcinoma with complete margin control: outcome at 5-year follow-up. Dermatology 2010;220:363–369. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20484877.

154. Schell AE, Russell MA, Park SS. Suggested excisional margins for cutaneous malignant lesions based on Mohs micrographic surgery. JAMA Facial Plast Surg 2013;15:337–343. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23744451.

155. Quazi SJ, Aslam N, Saleem H, et al. Surgical margin of excision in basal cell carcinoma: A systematic review of literature. Cureus 2020;12:e9211. Available at: https://www.ncbi.nlm.nih.gov/pubmed/32821563.

156. Bisson MA, Dunkin CS, Suvarna SK, Griffiths RW. Do plastic surgeons resect basal cell carcinomas too widely? A prospective study comparing surgical and histological margins. Br J Plast Surg 2002;55:293–297. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12160534.

157. Gulleth Y, Goldberg N, Silverman RP, Gastman BR. What is the best surgical margin for a Basal cell carcinoma: a meta-analysis of the literature. Plast Reconstr Surg 2010;126:1222–1231. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20885244.

158. Thomas DJ, King AR, Peat BG. Excision margins for nonmelanotic skin cancer. Plast Reconstr Surg 2003;112:57–63. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12832877.

159. Fraga SD, Besaw RJ, Murad F, et al. Complete margin assessment versus sectional assessment in surgically excised high-risk keratinocyte carcinomas: A systematic review and meta-analysis. Dermatol Surg 2022;48:704–710. Available at: https://www.ncbi.nlm.nih.gov/pubmed/35778249.

160. Su SY, Giorlando F, Ek EW, Dieu T. Incomplete excision of basal cell carcinoma: a prospective trial. Plast Reconstr Surg 2007;120:1240–1248. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17898596.

161. Farhi D, Dupin N, Palangie A, et al. Incomplete excision of basal cell carcinoma: rate and associated factors among 362 consecutive cases. Dermatol Surg 2007;33:1207–1214. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17903153.

162. Kappelin J, Nielsen K, Nilsson F, et al. Surgical treatment of basal cell carcinoma: a case series on factors influencing the risk of an incomplete primary excision. J Eur Acad Dermatol Venereol 2020;34:2518–2525. Available at: https://www.ncbi.nlm.nih.gov/pubmed/32124503.

163. Dieu T, Macleod AM. Incomplete excision of basal cell carcinomas: a retrospective audit. ANZ J Surg 2002;72:219–221. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12071456.

164. Ramdas K, van Lee C, Beck S, et al. Differences in rate of complete excision of basal cell carcinoma by dermatologists, plastic surgeons and general practitioners: A large cross-sectional study. Dermatology 2018;234:86–91. Available at: https://www.ncbi.nlm.nih.gov/pubmed/30086541.

165. Sexton M, Jones DB, Maloney ME. Histologic pattern analysis of basal cell carcinoma. Study of a series of 1039 consecutive neoplasms. J Am Acad Dermatol 1990;23:1118–1126. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2273112.

166. Sherry KR, Reid LA, Wilmshurst AD. A five year review of basal cell carcinoma excisions. J Plast Reconstr Aesthet Surg 2010;63:1485–1489. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19837017.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

167. Griffiths RW, Suvarna SK, Stone J. Basal cell carcinoma histological clearance margins: an analysis of 1539 conventionally excised tumours. Wider still and deeper? J Plast Reconstr Aesthet Surg 2007;60:41–47. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17126265.

168. Masud D, Moustaki M, Staruch R, Dheansa B. Basal cell carcinomata: Risk factors for incomplete excision and results of re-excision. J Plast Reconstr Aesthet Surg 2016;69:652–656. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26948998.

169. Malik V, Goh KS, Leong S, et al. Risk and outcome analysis of 1832 consecutively excised basal cell carcinomas in a tertiary referral plastic surgery unit. J Plast Reconstr Aesthet Surg 2010;63:2057–2063. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20226750.

170. Wolf DJ, Zitelli JA. Surgical margins for basal cell carcinoma. Arch Dermatol 1987;123:340–344. Available at: https://www.ncbi.nlm.nih.gov/pubmed/3813602.

171. Muller FM, Dawe RS, Moseley H, Fleming CJ. Randomized comparison of Mohs micrographic surgery and surgical excision for small nodular basal cell carcinoma: tissue-sparing outcome. Dermatol Surg 2009;35:1349–1354. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19500127.

172. Gniadecki R, Glud M, Mortensen K, et al. Favourable results of Mohs micrographic surgery for basal cell carcinoma. Dan Med J 2015;62:A5171. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26621396.

173. Curtis KK, Fakult NJ, Strunck JL, et al. Establishing Consensus for Mohs Micrographic Surgical Techniques in the Treatment of Melanoma in Situ for Future Clinical Trials: A Modified Delphi Study. J Natl Compr Canc Netw 2024:1–6. Available at: https://www.ncbi.nlm.nih.gov/pubmed/39079545.

174. Weesie F, Naus NC, Vasilic D, et al. Recurrence of periocular basal cell carcinoma and squamous cell carcinoma after Mohs micrographic surgery: a retrospective cohort study. Br J Dermatol 2019;180:1176–1182. Available at: https://www.ncbi.nlm.nih.gov/pubmed/30536656.

175. Drucker AM, Adam GP, Rofeberg V, et al. Treatments of primary basal cell carcinoma of the skin: A systematic review and network meta-analysis. Ann Intern Med 2018;169:456–466. Available at: https://www.ncbi.nlm.nih.gov/pubmed/30242379.

176. Wennberg AM, Larko O, Stenquist B. Five-year results of Mohs' micrographic surgery for aggressive facial basal cell carcinoma in Sweden. Acta Derm Venereol 1999;79:370–372. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10494714.

177. Veronese F, Farinelli P, Zavattaro E, et al. Basal cell carcinoma of the head region: therapeutical results of 350 lesions treated with Mohs micrographic surgery. J Eur Acad Dermatol Venereol 2012;26:838–843. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21707774.

178. Litwin AS, Rytina E, Ha T, et al. Management of periocular basal cell carcinoma by Mohs micrographic surgery. J Dermatolog Treat 2013;24:232–234. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22551161.

179. Sin CW, Barua A, Cook A. Recurrence rates of periocular basal cell carcinoma following Mohs micrographic surgery: a retrospective study. Int J Dermatol 2016;55:1044–1047. Available at: https://www.ncbi.nlm.nih.gov/pubmed/27152747.

180. Leibovitch I, Huilgol SC, Selva D, et al. Basal cell carcinoma treated with Mohs surgery in Australia II. Outcome at 5-year follow-up. J Am Acad Dermatol 2005;53:452–457. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16112352.

181. Malhotra R, Huilgol SC, Huynh NT, Selva D. The Australian Mohs database, part II: periocular basal cell carcinoma outcome at 5-year follow-up. Ophthalmology 2004;111:631–636. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15051193.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

182. Mosterd K, Krekels GA, Nieman FH, et al. Surgical excision versus Mohs' micrographic surgery for primary and recurrent basal-cell carcinoma of the face: a prospective randomised controlled trial with 5-years' follow-up. Lancet Oncol 2008;9:1149–1156. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19010733.

183. Smeets NW, Krekels GA, Ostertag JU, et al. Surgical excision vs Mohs' micrographic surgery for basal-cell carcinoma of the face: randomised controlled trial. Lancet 2004;364:1766–1772. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15541449.

184. Flohil SC, van Dorst AM, Nijsten T, et al. Mohs micrographic surgery for basal cell carcinomas: appropriateness of 'Rotterdam' criteria and predictive factors for three or more stages. J Eur Acad Dermatol Venereol 2013;27:1228–1235. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23039378.

185. Takata Pontes L, Fantelli Stelini R, Cintra ML, et al. The importance of superficial basal cell carcinoma in a retrospective study of 139 patients who underwent Mohs micrographic surgery in a Brazilian university hospital. Clinics (Sao Paulo) 2015;70:721–725. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26602517.

186. Morris DS, Elzaridi E, Clarke L, et al. Periocular basal cell carcinoma: 5-year outcome following Slow Mohs surgery with formalin-fixed paraffin-embedded sections and delayed closure. Br J Ophthalmol 2009;93:474–476. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19060015.

187. Mendenhall WM, Amdur RJ, Hinerman RW, et al. Radiotherapy for cutaneous squamous and basal cell carcinomas of the head and neck. Laryngoscope 2009;119:1994–1999. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19688856.

188. Mendenhall WM, Ferlito A, Takes RP, et al. Cutaneous head and neck basal and squamous cell carcinomas with perineural invasion. Oral Oncol 2012;48:918–922. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22425152.

189. Han A, Ratner D. What is the role of adjuvant radiotherapy in the treatment of cutaneous squamous cell carcinoma with perineural invasion? Cancer 2007;109:1053–1059. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17279578.

190. Wilder RB, Kittelson JM, Shimm DS. Basal cell carcinoma treated with radiation therapy. Cancer 1991;68:2134–2137. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1913451.

191. Wilder RB, Shimm DS, Kittelson JM, et al. Recurrent basal cell carcinoma treated with radiation therapy. Arch Dermatol 1991;127:1668–1672. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1952970.

192. Childers BJ, Goldwyn RM, Ramos D, et al. Long-term results of irradiation for basal cell carcinoma of the skin of the nose. Plast Reconstr Surg 1994;93:1169–1173. Available at: https://www.ncbi.nlm.nih.gov/pubmed/8171136.

193. Hernandez-Machin B, Borrego L, Gil-Garcia M, Hernandez BH. Office-based radiation therapy for cutaneous carcinoma: evaluation of 710 treatments. Int J Dermatol 2007;46:453–459. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17472670.

194. Silverman MK, Kopf AW, Gladstein AH, et al. Recurrence rates of treated basal cell carcinomas. Part 4: X-ray therapy. J Dermatol Surg Oncol 1992;18:549–554. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1624628.

195. Zagrodnik B, Kempf W, Seifert B, et al. Superficial radiotherapy for patients with basal cell carcinoma: recurrence rates, histologic subtypes, and expression of p53 and Bcl-2. Cancer 2003;98:2708–2714. Available at: https://www.ncbi.nlm.nih.gov/pubmed/14669293.

196. Cognetta AB, Howard BM, Heaton HP, et al. Superficial x-ray in the treatment of basal and squamous cell carcinomas: a viable option in select patients. J Am Acad Dermatol 2012;67:1235–1241. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22818756.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

197. Avril MF, Auperin A, Margulis A, et al. Basal cell carcinoma of the face: surgery or radiotherapy? Results of a randomized study. Br J Cancer 1997;76:100–106. Available at: https://www.ncbi.nlm.nih.gov/pubmed/9218740.

198. Petit JY, Avril MF, Margulis A, et al. Evaluation of cosmetic results of a randomized trial comparing surgery and radiotherapy in the treatment of basal cell carcinoma of the face. Plast Reconstr Surg 2000;105:2544–2551. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10845311.

199. Hall VL, Leppard BJ, McGill J, et al. Treatment of basal-cell carcinoma: comparison of radiotherapy and cryotherapy. Clin Radiol 1986;37:33–34. Available at: https://www.ncbi.nlm.nih.gov/pubmed/3514075.

200. Garcia-Martin E, Gil-Arribas LM, Idoipe M, et al. Comparison of imiquimod 5% cream versus radiotherapy as treatment for eyelid basal cell carcinoma. Br J Ophthalmol 2011;95:1393–1396. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21242584.

201. Zaorsky NG, Lee CT, Zhang E, et al. Hypofractionated radiation therapy for basal and squamous cell skin cancer: A meta-analysis. Radiother Oncol 2017;125:13–20. Available at: https://www.ncbi.nlm.nih.gov/pubmed/28843727.

202. van Hezewijk M, Creutzberg CL, Putter H, et al. Efficacy of a hypofractionated schedule in electron beam radiotherapy for epithelial skin cancer: Analysis of 434 cases. Radiother Oncol 2010;95:245–249. Available at: http://www.ncbi.nlm.nih.gov/pubmed/20334941.

203. Hu M, Liu H, Bernard A, et al. Hypofractionated adjuvant radiotherapy in cutaneous squamous cell and basal cell carcinoma of the head and neck: 50(Gy) in 20 study. Oral Oncology Reports 2025;13:100732. Available at: https://www.sciencedirect.com/science/article/pii/S2772906025000202.

204. Likhacheva AO, Devlin PM, Shirvani SM, et al. Skin surface brachytherapy: A survey of contemporary practice patterns. Brachytherapy 2017;16:223–229. Available at: https://www.ncbi.nlm.nih.gov/pubmed/27908679.

205. Ashby MA, Pacella JA, de Groot R, Ainslie J. Use of a radon mould technique for skin cancer: results from the Peter MacCallum Cancer Institute (1975-1984). Br J Radiol 1989;62:608–612. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2758248.

206. Cipriani C, Desantis M, Dahlhoff G, et al. Personalized irradiation therapy for NMSC by rhenium-188 skin cancer therapy: a long-term retrospective study. J Dermatolog Treat 2022;33:969–975. Available at: https://www.ncbi.nlm.nih.gov/pubmed/32648530.

207. Sedda AF, Rossi G, Cipriani C, et al. Dermatological high-dose-rate brachytherapy for the treatment of basal and squamous cell carcinoma. Clin Exp Dermatol 2008;33:745–749. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18681873.

208. Guibert M, David I, Vergez S, et al. Brachytherapy in lip carcinoma: long-term results. Int J Radiat Oncol Biol Phys 2011;81:e839–843. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21163589.

209. Gauden R, Pracy M, Avery AM, et al. HDR brachytherapy for superficial non-melanoma skin cancers. J Med Imaging Radiat Oncol 2013;57:212–217. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23551783.

210. Ballester-Sanchez R, Pons-Llanas O, Candela-Juan C, et al. Two years results of electronic brachytherapy for basal cell carcinoma. J Contemp Brachytherapy 2017;9:251–255. Available at: https://www.ncbi.nlm.nih.gov/pubmed/28725249.

211. Bhatnagar A, Loper A. The initial experience of electronic brachytherapy for the treatment of non-melanoma skin cancer. Radiat Oncol 2010;5:87. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20875139.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

212. Eaton DJ. Electronic brachytherapy--current status and future directions. Br J Radiol 2015;88:20150002. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25748070.

213. Tom MC, Hepel JT, Patel R, et al. The american brachytherapy society consensus statement for electronic brachytherapy. Brachytherapy 2019;18:292–298. Available at: https://www.ncbi.nlm.nih.gov/pubmed/30497939.

214. Braathen LR, Szeimies RM, Basset-Seguin N, et al. Guidelines on the use of photodynamic therapy for nonmelanoma skin cancer: an international consensus. International Society for Photodynamic Therapy in Dermatology, 2005. J Am Acad Dermatol 2007;56:125–143. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17190630.

215. Jansen MHE, Mosterd K, Arits A, et al. Five-year results of a randomized controlled trial comparing effectiveness of photodynamic therapy, topical imiquimod, and topical 5-fluorouracil in patients with superficial basal cell carcinoma. J Invest Dermatol 2018;138:527–533. Available at: https://www.ncbi.nlm.nih.gov/pubmed/29045820.

216. Drew BA, Karia PS, Mora AN, et al. Treatment patterns, outcomes, and patient satisfaction of primary epidermally limited nonmelanoma skin cancer. Dermatol Surg 2017;43:1423–1430. Available at: https://www.ncbi.nlm.nih.gov/pubmed/28661992.

217. Backman EJ, Polesie S, Berglund S, et al. Curettage vs. cryosurgery for superficial basal cell carcinoma: a prospective, randomised and controlled trial. J Eur Acad Dermatol Venereol 2022;36:1758–1765. Available at: https://www.ncbi.nlm.nih.gov/pubmed/35543079.

218. Loranger N, Mirza FN, Lee T, et al. Combined topical 5-fluorouracil and calcipotriene effectively treats superficial keratinocyte carcinomas: A retrospective cohort study from 2 academic centers. J Am Acad Dermatol 2025;92:610–612. Available at: https://www.ncbi.nlm.nih.gov/pubmed/39532231.

219. Peris K, Campione E, Micantonio T, et al. Imiquimod treatment of superficial and nodular basal cell carcinoma: 12-week open-label trial. Dermatol Surg 2005;31:318–323. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15841634.

220. Eigentler TK, Kamin A, Weide BM, et al. A phase III, randomized, open label study to evaluate the safety and efficacy of imiquimod 5% cream applied thrice weekly for 8 and 12 weeks in the treatment of low-risk nodular basal cell carcinoma. J Am Acad Dermatol 2007;57:616–621. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17610993.

221. Geisse J, Caro I, Lindholm J, et al. Imiquimod 5% cream for the treatment of superficial basal cell carcinoma: results from two phase III, randomized, vehicle-controlled studies. J Am Acad Dermatol 2004;50:722–733. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15097956.

222. Schulze HJ, Cribier B, Requena L, et al. Imiquimod 5% cream for the treatment of superficial basal cell carcinoma: results from a randomized vehicle-controlled phase III study in Europe. Br J Dermatol 2005;152:939–947. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15888150.

223. Quirk C, Gebauer K, De'Ambrosis B, et al. Sustained clearance of superficial basal cell carcinomas treated with imiquimod cream 5%: results of a prospective 5-year study. Cutis 2010;85:318–324. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20666194.

224. Marks R, Gebauer K, Shumack S, et al. Imiquimod 5% cream in the treatment of superficial basal cell carcinoma: results of a multicenter 6-week dose-response trial. J Am Acad Dermatol 2001;44:807–813. Available at: https://www.ncbi.nlm.nih.gov/pubmed/11312429.

225. Gollnick H, Barona CG, Frank RG, et al. Recurrence rate of superficial basal cell carcinoma following treatment with imiquimod 5% cream: conclusion of a 5-year long-term follow-up study in Europe. Eur J Dermatol 2008;18:677–682. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18955210.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

226. McKay KM, Sambrano BL, Fox PS, et al. Thickness of superficial basal cell carcinoma (sBCC) predicts imiquimod efficacy: a proposal for a thickness-based definition of sBCC. Br J Dermatol 2013;169:549–554. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23627639.

227. Bath-Hextall F, Ozolins M, Armstrong SJ, et al. Surgical excision versus imiquimod 5% cream for nodular and superficial basal-cell carcinoma (SINS): a multicentre, non-inferiority, randomised controlled trial. Lancet Oncol 2014;15:96–105. Available at: https://www.ncbi.nlm.nih.gov/pubmed/24332516.

228. Williams HC, Bath-Hextall F, Ozolins M, et al. Surgery versus 5% imiquimod for nodular and superficial basal cell carcinoma: 5-year results of the sins randomized controlled trial. J Invest Dermatol 2017;137:614–619. Available at: https://www.ncbi.nlm.nih.gov/pubmed/27932240.

229. Metterle L, Nelson C, Patel N. Intralesional 5-fluorouracil (FU) as a treatment for nonmelanoma skin cancer (NMSC): A review. J Am Acad Dermatol 2016;74:552–557. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26577512.

230. Neale H, Michelon M, Jacob S, et al. Topical 5% 5-fluorouracil versus procedural modalities for squamous cell carcinoma in situ and superficial basal cell carcinoma: A retrospective cohort analysis. J Am Acad Dermatol 2022;87:423–425. Available at: https://www.ncbi.nlm.nih.gov/pubmed/34478783.

231. Arits AH, Mosterd K, Essers BA, et al. Photodynamic therapy versus topical imiquimod versus topical fluorouracil for treatment of superficial basal-cell carcinoma: a single blind, non-inferiority, randomised controlled trial. Lancet Oncol 2013;14:647–654. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23683751.

232. Roozeboom MH, Arits A, Mosterd K, et al. Three-year follow-up results of photodynamic therapy vs. Imiquimod vs. Fluorouracil for treatment of superficial basal cell carcinoma: A single-blind, noninferiority, randomized controlled trial. J Invest Dermatol 2016;136:1568–1574. Available at: https://www.ncbi.nlm.nih.gov/pubmed/27113429.

233. Gross K, Kircik L, Kricorian G. 5% 5-Fluorouracil cream for the treatment of small superficial Basal cell carcinoma: efficacy, tolerability, cosmetic outcome, and patient satisfaction. Dermatol Surg 2007;33:433–439; discussion 440. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17430377.

234. Reymann F. Treatment of basal cell carcinoma of the skin with 5-fluorouracil ointment. A 10-year follow-up study. Dermatologica 1979;158:368–372. Available at: https://www.ncbi.nlm.nih.gov/pubmed/437226.

235. Love WE, Bernhard JD, Bordeaux JS. Topical imiquimod or fluorouracil therapy for basal and squamous cell carcinoma: a systematic review. Arch Dermatol 2009;145:1431–1438. Available at: https://www.ncbi.nlm.nih.gov/pubmed/20026854.

236. Kuijpers DI, Thissen MR, Thissen CA, Neumann MH. Similar effectiveness of methyl aminolevulinate and 5-aminolevulinate in topical photodynamic therapy for nodular basal cell carcinoma. J Drugs Dermatol 2006;5:642–645. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16865869.

237. Savoia P, Deboli T, Previgliano A, Broganelli P. Usefulness of photodynamic therapy as a possible therapeutic alternative in the treatment of basal cell carcinoma. Int J Mol Sci 2015;16:23300–23317. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26426005.

238. Oseroff AR, Blumenson LR, Wilson BD, et al. A dose ranging study of photodynamic therapy with porfimer sodium (Photofrin) for treatment of basal cell carcinoma. Lasers Surg Med 2006;38:417–426. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16788928.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

239. Berroeta L, Clark C, Dawe RS, et al. A randomized study of minimal curettage followed by topical photodynamic therapy compared with surgical excision for low-risk nodular basal cell carcinoma. Br J Dermatol 2007;157:401–403. Available at: https://www.ncbi.nlm.nih.gov/pubmed/17573890.

240. Szeimies RM, Ibbotson S, Murrell DF, et al. A clinical study comparing methyl aminolevulinate photodynamic therapy and surgery in small superficial basal cell carcinoma (8-20 mm), with a 12-month follow-up. J Eur Acad Dermatol Venereol 2008;22:1302–1311. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18624836.

241. Roozeboom MH, Aardoom MA, Nelemans PJ, et al. Fractionated 5-aminolevulinic acid photodynamic therapy after partial debulking versus surgical excision for nodular basal cell carcinoma: a randomized controlled trial with at least 5-year follow-up. J Am Acad Dermatol 2013;69:280–287. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23566914.

242. Cosgarea R, Susan M, Crisan M, Senila S. Photodynamic therapy using topical 5-aminolaevulinic acid vs. surgery for basal cell carcinoma. J Eur Acad Dermatol Venereol 2013;27:980–984. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22738399.

243. Wang H, Xu Y, Shi J, et al. Photodynamic therapy in the treatment of basal cell carcinoma: a systematic review and meta-analysis. Photodermatol Photoimmunol Photomed 2015;31:44–53. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25377432.

244. Rhodes LE, de Rie M, Enstrom Y, et al. Photodynamic therapy using topical methyl aminolevulinate vs surgery for nodular basal cell carcinoma: results of a multicenter randomized prospective trial. Arch Dermatol 2004;140:17–23. Available at: https://www.ncbi.nlm.nih.gov/pubmed/14732655.

245. Collier NJ, Haylett AK, Wong TH, et al. Conventional and combination topical photodynamic therapy for basal cell carcinoma: systematic review and meta-analysis. Br J Dermatol 2018;179:1277–1296. Available at: https://www.ncbi.nlm.nih.gov/pubmed/29889302.

246. Zou Y, Zhao Y, Yu J, et al. Photodynamic therapy versus surgical excision to basal cell carcinoma: meta-analysis. J Cosmet Dermatol 2016;15:374–382. Available at: https://www.ncbi.nlm.nih.gov/pubmed/27363535.

247. Horn M, Wolf P, Wulf HC, et al. Topical methyl aminolaevulinate photodynamic therapy in patients with basal cell carcinoma prone to complications and poor cosmetic outcome with conventional treatment. Br J Dermatol 2003;149:1242–1249. Available at: https://www.ncbi.nlm.nih.gov/pubmed/14674903.

248. Christensen E, Mork C, Skogvoll E. High and sustained efficacy after two sessions of topical 5-aminolaevulinic acid photodynamic therapy for basal cell carcinoma: a prospective, clinical and histological 10-year follow-up study. Br J Dermatol 2012;166:1342–1348. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22309486.

249. Zeitouni NC, Shieh S, Oseroff AR. Laser and photodynamic therapy in the management of cutaneous malignancies. Clin Dermatol 2001;19:328–338. Available at: https://www.ncbi.nlm.nih.gov/pubmed/11479045.

250. Osiecka B, Jurczyszyn K, Ziolkowski P. The application of Levulan-based photodynamic therapy with imiquimod in the treatment of recurrent basal cell carcinoma. Med Sci Monit 2012;18:PI5–9. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22293891.

251. Mosterd K, Thissen MR, Nelemans P, et al. Fractionated 5-aminolaevulinic acid-photodynamic therapy vs. surgical excision in the treatment of nodular basal cell carcinoma: results of a randomized controlled trial. Br J Dermatol 2008;159:864–870. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18717680.

252. Fantini F, Greco A, Del Giovane C, et al. Photodynamic therapy for basal cell carcinoma: clinical and pathological determinants of response. J Eur Acad Dermatol Venereol 2011;25:896–901. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21054566.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

253. Vinciullo C, Elliott T, Francis D, et al. Photodynamic therapy with topical methyl aminolaevulinate for 'difficult-to-treat' basal cell carcinoma. Br J Dermatol 2005;152:765–772. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15840111.

254. Sotiriou E, Apalla Z, Maliamani F, et al. Intraindividual, right-left comparison of topical 5-aminolevulinic acid photodynamic therapy vs. 5% imiquimod cream for actinic keratoses on the upper extremities. J Eur Acad Dermatol Venereol 2009;23:1061–1065. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19470041.

255. Morton CA, McKenna KE, Rhodes LE, et al. Guidelines for topical photodynamic therapy: update. Br J Dermatol 2008;159:1245–1266. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18945319.

256. Kokoszka A, Scheinfeld N. Evidence-based review of the use of cryosurgery in treatment of basal cell carcinoma. Dermatol Surg 2003;29:566–571. Available at: https://www.ncbi.nlm.nih.gov/pubmed/12786697.

257. Kuflik EG, Gage AA. The five-year cure rate achieved by cryosurgery for skin cancer. J Am Acad Dermatol 1991;24:1002–1004. Available at: https://www.ncbi.nlm.nih.gov/pubmed/1820761.

258. Mallon E, Dawber R. Cryosurgery in the treatment of basal cell carcinoma. Assessment of one and two freeze-thaw cycle schedules. Dermatol Surg 1996;22:854–858. Available at: https://www.ncbi.nlm.nih.gov/pubmed/9246168.

259. Wang I, Bendsoe N, Klinteberg CA, et al. Photodynamic therapy vs. cryosurgery of basal cell carcinomas: results of a phase III clinical trial. Br J Dermatol 2001;144:832–840. Available at: https://www.ncbi.nlm.nih.gov/pubmed/11298545.

260. Basset-Seguin N, Ibbotson SH, Emtestam L, et al. Topical methyl aminolaevulinate photodynamic therapy versus cryotherapy for superficial basal cell carcinoma: a 5 year randomized trial. Eur J Dermatol 2008;18:547–553. Available at: https://www.ncbi.nlm.nih.gov/pubmed/18693158.

261. Thissen MR, Nieman FH, Ideler AH, et al. Cosmetic results of cryosurgery versus surgical excision for primary uncomplicated basal cell carcinomas of the head and neck. Dermatol Surg 2000;26:759–764. Available at: https://www.ncbi.nlm.nih.gov/pubmed/10940063.

262. Roozeboom MH, Arits AH, Nelemans PJ, Kelleners-Smeets NW. Overall treatment success after treatment of primary superficial basal cell carcinoma: a systematic review and meta-analysis of randomized and nonrandomized trials. Br J Dermatol 2012;167:733–756. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22612571.

263. Roozeboom MH, Nelemans PJ, Mosterd K, et al. Photodynamic therapy vs. topical imiquimod for treatment of superficial basal cell carcinoma: a subgroup analysis within a noninferiority randomized controlled trial. Br J Dermatol 2015;172:739–745. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25066012.

264. Chen AC, Martin AJ, Choy B, et al. A phase 3 randomized trial of nicotinamide for skin-cancer chemoprevention. N Engl J Med 2015;373:1618–1626. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26488693.

265. Chen AC, Martin AJ, Dalziell RA, et al. A phase II randomized controlled trial of nicotinamide for skin cancer chemoprevention in renal transplant recipients. Br J Dermatol 2016;175:1073–1075. Available at: https://www.ncbi.nlm.nih.gov/pubmed/27061568.

266. Mainville L, Smilga AS, Fortin PR. Effect of Nicotinamide in Skin Cancer and Actinic Keratoses Chemoprophylaxis, and Adverse Effects Related to Nicotinamide: A Systematic Review and Meta-Analysis. J Cutan Med Surg 2022;26:297–308. Available at: https://www.ncbi.nlm.nih.gov/pubmed/35134311.

267. Allen NC, Martin AJ, Snaidr VA, et al. Nicotinamide for Skin-Cancer Chemoprevention in Transplant Recipients. N Engl J Med 2023;388:804–812. Available at: https://www.ncbi.nlm.nih.gov/pubmed/36856616.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

268. Elmets CA, Viner JL, Pentland AP, et al. Chemoprevention of nonmelanoma skin cancer with celecoxib: a randomized, double-blind, placebo-controlled trial. J Natl Cancer Inst 2010;102:1835–1844. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21115882.

269. Kadakia KC, Barton DL, Loprinzi CL, et al. Randomized controlled trial of acitretin versus placebo in patients at high-risk for basal cell or squamous cell carcinoma of the skin (North Central Cancer Treatment Group Study 969251). Cancer 2012;118:2128–2137. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21882176.

270. Schauder DM, Kim J, Nijhawan RI. Evaluation of the use of capecitabine for the treatment and prevention of actinic keratoses, squamous cell carcinoma, and basal cell carcinoma: A systematic review. JAMA Dermatol 2020;156:1117–1124. Available at: https://www.ncbi.nlm.nih.gov/pubmed/32639538.

271. Tang JY, Chiou AS, Mackay-Wiggan JM, et al. Tazarotene: randomized, double-blind, vehicle-controlled, and open-label concurrent trials for basal cell carcinoma prevention and therapy in patients with basal cell nevus syndrome. Cancer Prev Res (Phila) 2014;7:292–299. Available at: https://www.ncbi.nlm.nih.gov/pubmed/24441673.

272. Drugs@FDA: FDA-Approved Drugs. U.S. Food & Drug Administration; Available at: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm. Accessed 02/10/2026.

273. Sekulic A, Migden MR, Oro AE, et al. Efficacy and safety of vismodegib in advanced basal-cell carcinoma. N Engl J Med 2012;366:2171–2179. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22670903.

274. Sekulic A, Migden MR, Lewis K, et al. Pivotal ERIVANCE basal cell carcinoma (BCC) study: 12-month update of efficacy and safety of vismodegib in advanced BCC. J Am Acad Dermatol 2015;72:1021–1026 e1028. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25981002.

275. Sekulic A, Migden MR, Basset-Seguin N, et al. Long-term safety and efficacy of vismodegib in patients with advanced basal cell carcinoma: final update of the pivotal ERIVANCE BCC study. BMC Cancer 2017;17:332. Available at: https://www.ncbi.nlm.nih.gov/pubmed/28511673.

276. Dreno B, Basset-Seguin N, Caro I, et al. Clinical benefit assessment of vismodegib therapy in patients with advanced basal cell carcinoma. Oncologist 2014;19:790–796. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25001266.

277. Barker CA, Dufault S, Arron ST, et al. Phase II, Single-Arm Trial of Induction and Concurrent Vismodegib With Curative-Intent Radiation Therapy for Locally Advanced, Unresectable Basal Cell Carcinoma. J Clin Oncol 2024;42:2327–2335. Available at: https://www.ncbi.nlm.nih.gov/pubmed/38630954.

278. Tang JY, Mackay-Wiggan JM, Aszterbaum M, et al. Inhibiting the hedgehog pathway in patients with the basal-cell nevus syndrome. N Engl J Med 2012;366:2180–2188. Available at: https://www.ncbi.nlm.nih.gov/pubmed/22670904.

279. Tang JY, Ally MS, Chanana AM, et al. Inhibition of the hedgehog pathway in patients with basal-cell nevus syndrome: final results from the multicentre, randomised, double-blind, placebo-controlled, phase 2 trial. Lancet Oncol 2016;17:1720–1731. Available at: https://www.ncbi.nlm.nih.gov/pubmed/27838224.

280. Ally MS, Tang JY, Joseph T, et al. The use of vismodegib to shrink keratocystic odontogenic tumors in patients with basal cell nevus syndrome. JAMA Dermatol 2014;150:542–545. Available at: https://www.ncbi.nlm.nih.gov/pubmed/24623282.

281. Migden MR, Guminski A, Gutzmer R, et al. Treatment with two different doses of sonidegib in patients with locally advanced or metastatic basal cell carcinoma (BOLT): a multicentre, randomised, double-blind phase 2 trial. Lancet Oncol 2015;16:716–728. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25981810.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

282. Lewis K, Dummer R, Farberg AS, et al. Effects of sonidegib following dose reduction and treatment interruption in patients with advanced basal cell carcinoma during 42-month BOLT trial. Dermatol Ther (Heidelb) 2021;11:2225–2234. Available at: https://www.ncbi.nlm.nih.gov/pubmed/34669179.

283. Lear JT, Migden MR, Lewis KD, et al. Long-term efficacy and safety of sonidegib in patients with locally advanced and metastatic basal cell carcinoma: 30-month analysis of the randomized phase 2 BOLT study. J Eur Acad Dermatol Venereol 2018;32:372–381. Available at: https://www.ncbi.nlm.nih.gov/pubmed/28846163.

284. Dummer R, Guminksi A, Gutzmer R, et al. Long-term efficacy and safety of sonidegib in patients with advanced basal cell carcinoma: 42-month analysis of the phase II randomized, double-blind BOLT study. Br J Dermatol 2020;182:1369–1378. Available at: https://www.ncbi.nlm.nih.gov/pubmed/31545507.

285. Dummer R, Lear JT, Guminski A, et al. Efficacy of sonidegib in histologic subtypes of advanced basal cell carcinoma: Results from the final analysis of the randomized phase 2 Basal Cell Carcinoma Outcomes With LDE225 Treatment (BOLT) trial at 42 months. J Am Acad Dermatol 2021;84:1162–1164. Available at: https://www.ncbi.nlm.nih.gov/pubmed/33358380.

286. Gutzmer R, Robert C, Loquai C, et al. Assessment of various efficacy outcomes using ERIVANCE-like criteria in patients with locally advanced basal cell carcinoma receiving sonidegib: results from a preplanned sensitivity analysis. BMC Cancer 2021;21:1244. Available at: https://www.ncbi.nlm.nih.gov/pubmed/34798846.

287. Danial C, Sarin KY, Oro AE, Chang AL. An investigator-initiated open-label trial of sonidegib in advanced basal cell carcinoma patients resistant to vismodegib. Clin Cancer Res 2016;22:1325–1329. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26546616.

288. Ally MS, Aasi S, Wysong A, et al. An investigator-initiated open-label clinical trial of vismodegib as a neoadjuvant to surgery for high-risk basal cell carcinoma. J Am Acad Dermatol 2014;71:904–911 e901. Available at: https://www.ncbi.nlm.nih.gov/pubmed/24929884.

289. Sofen H, Gross KG, Goldberg LH, et al. A phase II, multicenter, open-label, 3-cohort trial evaluating the efficacy and safety of vismodegib in operable basal cell carcinoma. J Am Acad Dermatol 2015;73:99–105 e101. Available at: https://www.ncbi.nlm.nih.gov/pubmed/25913533.

290. Tauber G, Pavlovsky L, Fenig E, Hodak E. Vismodegib for radiation-induced multiple basal cell carcinomas (BCCs) of the scalp. J Am Acad Dermatol 2015;73:799–801. Available at: https://www.ncbi.nlm.nih.gov/pubmed/26320385.

291. Bertrand N, Guerreschi P, Basset-Seguin N, et al. Vismodegib in neoadjuvant treatment of locally advanced basal cell carcinoma: First results of a multicenter, open-label, phase 2 trial (VISMONEO study): Neoadjuvant Vismodegib in Locally Advanced Basal Cell Carcinoma. EClinicalMedicine 2021;35:100844. Available at: https://www.ncbi.nlm.nih.gov/pubmed/33997740.

292. Dreno B, Kunstfeld R, Hauschild A, et al. Two intermittent vismodegib dosing regimens in patients with multiple basal-cell carcinomas (MIKIE): a randomised, regimen-controlled, double-blind, phase 2 trial. Lancet Oncol 2017;18:404–412. Available at: https://www.ncbi.nlm.nih.gov/pubmed/28188086.

293. Scalvenzi M, Cappello M, Costa C, et al. Low-dose vismodegib as maintenance therapy after locally advanced basal cell carcinoma complete remission: High efficacy with minimal toxicity. Dermatol Ther (Heidelb) 2020;10:465–468. Available at: https://www.ncbi.nlm.nih.gov/pubmed/32240528.

294. Jimeno A, Weiss GJ, Miller WH, Jr., et al. Phase I study of the hedgehog pathway inhibitor IPI-926 in adult patients with solid tumors. Clin Cancer Res 2013;19:2766–2774. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23575478.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.

PLEASE NOTE that use of this NCCN Content is governed by the End-User License Agreement, and you MAY NOT distribute this Content or use it with any artificial intelligence model or tool.
Printed by Máximo López Chenlo on 5/4/2026 6:13:11 PM. Copyright © 2026 National Comprehensive Cancer Network, Inc. All Rights Reserved.

295. Kim DJ, Kim J, Spaunhurst K, et al. Open-label, exploratory phase II trial of oral itraconazole for the treatment of basal cell carcinoma. J Clin Oncol 2014;32:745–751. Available at: https://www.ncbi.nlm.nih.gov/pubmed/24493717.

296. Stratigos AJ, Sekulic A, Peris K, et al. Cemiplimab in locally advanced basal cell carcinoma after hedgehog inhibitor therapy: an open-label, multi-centre, single-arm, phase 2 trial. Lancet Oncol 2021;22:848–857. Available at: https://www.ncbi.nlm.nih.gov/pubmed/34000246.

297. Stratigos AJ, Chen CI, Ivanescu C, et al. Quality of life in cemiplimab-treated patients with locally advanced basal cell carcinoma in a Phase II clinical trial. Future Oncol 2024;20:2249–2258. Available at: https://www.ncbi.nlm.nih.gov/pubmed/39073799.

298. Lewis KD, Peris K, Sekulic A, et al. Final analysis of phase II results with cemiplimab in metastatic basal cell carcinoma after hedgehog pathway inhibitors. Ann Oncol 2024;35:221–228. Available at: https://www.ncbi.nlm.nih.gov/pubmed/38072158.

299. Carneiro BA, Watkin WG, Mehta UK, Brockstein BE. Metastatic basal cell carcinoma: complete response to chemotherapy and associated pure red cell aplasia. Cancer Invest 2006;24:396–400. Available at: https://www.ncbi.nlm.nih.gov/pubmed/16777692.

300. Moeholt K, Aagaard H, Pfeiffer P, Hansen O. Platinum-based cytotoxic therapy in basal cell carcinoma--a review of the literature. Acta Oncol 1996;35:677–682. Available at: https://www.ncbi.nlm.nih.gov/pubmed/8938213.

301. Ganti AK, Kessinger A. Systemic therapy for disseminated basal cell carcinoma: an uncommon manifestation of a common cancer. Cancer Treat Rev 2011;37:440–443. Available at: https://www.ncbi.nlm.nih.gov/pubmed/21216106.

302. Jefford M, Kiffer JD, Somers G, et al. Metastatic basal cell carcinoma: rapid symptomatic response to cisplatin and paclitaxel. ANZ J Surg 2004;74:704–705. Available at: https://www.ncbi.nlm.nih.gov/pubmed/15315581.

303. Wysong A, Aasi SZ, Tang JY. Update on metastatic basal cell carcinoma: a summary of published cases from 1981 through 2011. JAMA Dermatol 2013;149:615–616. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23677097.

304. Pfeiffer P, Hansen O, Rose C. Systemic cytotoxic therapy of basal cell carcinoma. A review of the literature. Eur J Cancer 1990;26:73–77. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2138485.

305. Guthrie TH, Jr., Porubsky ES, Luxenberg MN, et al. Cisplatin-based chemotherapy in advanced basal and squamous cell carcinomas of the skin: results in 28 patients including 13 patients receiving multimodality therapy. J Clin Oncol 1990;8:342–346. Available at: https://www.ncbi.nlm.nih.gov/pubmed/2405109.

306. Robinson JK. Follow-up and prevention (basal cell carcinoma). In: Miller SJ, Maloney ME, eds. Cutaneous Oncology Pathophysiology, diagnosis, and management. Malden, MA: Blackwell Science; 1998:695–698.

307. Marcil I, Stern RS. Risk of developing a subsequent nonmelanoma skin cancer in patients with a history of nonmelanoma skin cancer: a critical review of the literature and meta-analysis. Arch Dermatol 2000;136:1524–1530. Available at: https://www.ncbi.nlm.nih.gov/pubmed/11115165.

308. Ramachandran S, Rajaratnam R, Smith AG, et al. Patients with both basal and squamous cell carcinomas are at a lower risk of further basal cell carcinomas than patients with only a basal cell carcinoma. J Am Acad Dermatol 2009;61:247–251. Available at: https://www.ncbi.nlm.nih.gov/pubmed/19481292.

309. Flohil SC, van der Leest RJ, Arends LR, et al. Risk of subsequent cutaneous malignancy in patients with prior keratinocyte carcinoma: a systematic review and meta-analysis. Eur J Cancer 2013;49:2365–2375. Available at: https://www.ncbi.nlm.nih.gov/pubmed/23608733.

Version 2.2025 © 2025 National Comprehensive Cancer Network© (NCCN©), All rights reserved. NCCN Guidelines® and this illustration may not be reproduced in any form without the express written permission of NCCN.